



Management Science

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<http://pubsonline.informs.org>

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To cite this article:

Ga Young Ko, Donghyuk Shin, Seigyoung Auh, Yeonjung Lee, Sang Pil Han (2023) Learning Outside the Classroom During a Pandemic: Evidence from an Artificial Intelligence-Based Education App. *Management Science* 69(6):3616–3649. <https://doi.org/10.1287/mnsc.2022.4531>

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Learning Outside the Classroom During a Pandemic: Evidence from an Artificial Intelligence-Based Education App

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Received: April 10, 2021

Revised: November 25, 2021; May 7, 2022;
June 23, 2022

Accepted: June 27, 2022

Published Online in Articles in Advance:
September 12, 2022

<https://doi.org/10.1287/mnsc.2022.4531>

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Abstract. Drawing on the notion of compensatory behavior, this paper studies how students compensate for learning loss during a pandemic and what role artificial intelligence (AI) plays in this regard. We further probe into a difference in compensatory behavior for learning loss in terms of quantity, pattern, and pace (i.e., tripartite aspect of learning behavior) of AI-powered learning app usage depending on the level of pandemic threat and the proximity of a goal to students. Results show that the pandemic threat affects student learning behavior differently. Immediately following the COVID-19 outbreak, students who live in the epicenter of the outbreak (versus those who do not) use the app less at first, but with time, they use it more (quantity), on a more regular basis (pattern), and rebound to a curriculum path (pace) comparable to students who do not live in the outbreak's epicenter. These findings collectively explain behavior that is consistent with compensation for learning loss. The results also partially corroborate the goal-proximity effect, revealing that proximity to a goal (e.g., the degree to which the national university admission exam is approaching) has a moderating role in explaining the tripartite perspective of student learning behavior. Overall, these findings have important theoretical and practical implications for understanding how innovative education technologies can not only facilitate student learning during adversity, but also support learning recovery after adversity.

History: Accepted by D. J. Wu, information systems.

Supplemental Material: Data files available at <https://doi.org/10.1287/mnsc.2022.4531>.

Keywords: artificial intelligence • education app • COVID-19 • learning loss • compensatory behavior

1. Introduction

The COVID-19 pandemic has led to tumultuous disruptions permeating nearly every aspect of our lives. Unlike other natural (Jia et al. 2021, Lu et al. 2012) or epidemiological disasters (Tausczik et al. 2012, Blakey and Abramowitz 2017, Thompson et al. 2017) that were confined to certain geographical locations or were short-lived, the sweeping and deleterious effects of COVID-19 have been widespread, have been felt in every country around the globe, and have left people uncertain about when normalcy might be restored.

Globally, one of the sectors that has been hit the hardest as a result of the pandemic is education (specifically, kindergarten through 12th grade (K-12)). The pandemic has further exacerbated income disparities in society (e.g., Buckley and Barua 2020 and Jurzyk et al. 2020) and has led to a drastic shift in consumer behavior (e.g., Cox et al. 2020 and Sheth 2020). Foremost on the education front, the pandemic has elevated the need for a transformation in current education models

from in-person to the use of remote (or hybrid) learning technologies. As a testament to this demand, the United Nations Educational, Scientific and Cultural Organization (UNESCO) Director-General, Audrey Azoulay, stated: “We are entering uncharted territory and working with countries to find hi-tech, low-tech and no-tech solutions to assure the continuity of learning” (UNESCO 2020). Clearly, a pressing challenge for the global education community is to minimize students’ loss of learning by offering an effective learning model that is not only technologically and economically feasible, but also one that students will adopt and use.

To this end, the purpose of the current paper is to address how the use of an Information and Communication Technology (ICT)¹ artifact—an artificial intelligence (AI hereinafter)-powered math education app—among K-12 students can contribute to addressing the learning disruption brought about by COVID-19. Consequently, the broad research question we ask in the paper is: “How do students compensate for

learning loss during a pandemic, and what is the role of AI in this regard?” To further fine-tune this broader inquiry, we also examine a more nuanced question: “Is there a difference in compensatory behavior for learning loss when using an AI-powered learning app in terms of quantity, pattern, and pace of usage depending on (a) the level of pandemic threat and (b) the proximity of goal to students?”

To date, research on the impact of COVID-19 on the education sector is still in its nascent stage and can be summarized according to three broad themes (see Table 1): (1) The gap in learning resulting from remote-learning models will increase the income inequality between low and high socioeconomic status (SES) students (e.g., Bacher-Hicks et al. 2021); (2) the learning loss as a consequence of missing school and not receiving in-person coaching and guidance from teachers leads to significant life-long learning loss, with a greater loss expected for students of color (e.g., Dorn et al. 2020); and (3) students’ search for online learning resources (e.g., Google Classroom²) and online practice tools (e.g., Renaissance’s Freckle Math³) has increased significantly (e.g., Wyse et al. 2020).

Although much attention has been devoted to examining learning and potential income loss due to the pandemic (e.g., Dorn et al. 2020), notably missing from the literature are recovery behaviors or actions undertaken by students to compensate for such learning loss. In this respect, we advance the extant literature forward and fill an important void by studying how students can use AI to help them compensate for learning loss due to the pandemic. Despite recent research, as revealed in Table 1, the extant literature on the impact of COVID-19 on student learning behavior and outcome is embryonic at best when it comes to how ICT artifacts such as AI-based education apps contribute to compensate for learning loss, especially at the individual level. In terms of compensatory behavior for learning loss due to the pandemic, the following is still unknown and has received scant empirical attention, upon which this paper is able to shed light: (a) how much compensatory behavior occurs (i.e., quantity), (b) how compensatory behavior unfolds (i.e., pattern), and (c) how quickly compensatory behavior occurs (i.e., pace). Our paper is able to demonstrate a novel and holistic perspective into students’ compensatory behavior induced by the use of an AI-based education app for learning loss during the pandemic by providing this tripartite information of quantity, pattern, and pace.

We provide some context to our study to put our research into perspective. Although most distant student learning, whether synchronous or asynchronous, was made possible through remote-learning technologies such as Google Classroom or Zoom⁴ when in-class education was not feasible, there is sparse empirical research on students’ use of supplementary

remote-learning technologies, such as Photomath⁵ or Microsoft Math Solver,⁶ that typically occur outside of the classroom for tutorial purposes. The distinction between the usage of school-driven versus supplementary remote-learning technology is important because, in the case of the school-driven remote-learning technology, students do not have a choice but to participate, as it is involuntary in nature (i.e., no way to opt out unless students want to forgo school education altogether). However, for the supplementary remote-learning technology, which is the focus of this paper, the choice of whether to use it or not is entirely voluntary. Hence, the volitional aspect of the supplementary remote-learning technology adds a novel perspective to the research, given the paucity of empirical studies.

Based on a thorough review of the literature, we have identified the following gaps, which this paper aims to fill. First, no research, to date, has studied how students’ use of AI-based education apps varies depending on whether they are near the epicenter of the outbreak or not. That is, no research has examined “pandemic threat”—measured by the proximity to the epicenter of the outbreak—which we argue is an important factor that drives differential compensatory behavior (i.e., the use of an AI-powered learning app) in terms of quantity, pattern, and pace over time between students who live in the outbreak versus those who do not. Second, although education inequality has been the focus of attention regarding how COVID-19 has differentially affected students from diverse socioeconomic backgrounds, no research has examined the role of goals. Here, we examine goal proximity (i.e., whether a goal is proximal or distal) as a moderator of the impact of the pandemic threat on the quantity, pattern, and pace of compensatory behavior across time. We argue that although a high pandemic threat may impede learning and subsequent recovery, the countervailing effect of goal proximity can mitigate its negative effects. Figure 1 illustrates our conceptual model.

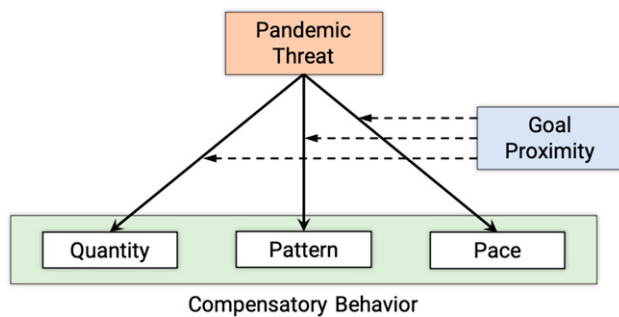
Against the above backdrop, we examine how South Korean K-12 students’ usage of QANDA, an AI-powered education app for math, changes over a 13-week period following the COVID-19 outbreak at the individual student level. QANDA facilitates remote learning by enabling students to receive solutions with line-by-line explanations by taking pictures of math problems and uploading them to the platform. We compare two groups of students’ (i.e., high school versus middle school) usage of QANDA between the metropolitan city of Daegu and the greater Gyeongbuk province (referred to as the DG region hereinafter)—the epicenter of the outbreak in South Korea—and non-DG areas.

Our research contributes to the information systems and education technology (EdTech) literatures in the following ways. First, we test how the pandemic threat affects the use of an AI-powered learning app over time for two disparate regions (i.e., DG versus

Table 1. Select Research on the Effect of COVID-19 on Education

Study	Key findings	Methodology/data	Context	Subject	Level of analysis
Bacher-Hicks et al. (2021)	The COVID-19 pandemic has led to an exacerbation of the gap between low- and high-SES students' access to educational resources.	Difference-in-difference using weekly Google Trends of search keywords related to online learning	K-12 education	Household (students and parents)	Aggregate
Dorn et al. (2020)	Loss of learning due to COVID-19 results in loss of earnings, but has caused disproportionately more learning loss for students of color and is expected to widen future earning disparity between different races in the United States.	Tabulation using multiple sources of data, such as from surveys, think tanks (e.g., Brookings Institute), and government agencies (e.g., Bureau of Labor Statistics)	K-12 education	Students	Aggregate (by race)
Wyse et al. (2020)	(1) The adverse impact of learning loss for math is more evenly distributed across grades K-12 than for reading, where the majority of damage to students is done in earlier grades (K-2). (2) The use of digital educational platforms for math increased after vs. before the COVID-19 outbreak to compensate for learning loss.	Tabulation using observational data from EdTech provider Renaissance	K-12 education	Students	Aggregate (by grade)
Andrew et al. (2020)	The learning loss (up to 35 days over a six-month period) is greater for less-affluent families.	Online survey	K-12 education	Parents	Individual
Lee and Steer (2019)	Integrating AI with human skills will allow scalable, yet personalized, education that will help maximize students' potential (e.g., Mindspark: Indian AI platform).	Conceptual	K-12 and university education	Not applicable	Not applicable
Prime et al. (2020)	The deleterious impact of COVID-19 on child adjustment works by infiltrating the family and breaking down and destabilizing the "3 R's" of the organization process of a family: routine, ritual, and rules.	Conceptual	Family	Not applicable	Not applicable
Current study	Empirically document compensatory behavior for learning loss from a tripartite perspective (i.e., quantity, pattern, and pace).	Difference-in-difference using primary and observational data of student usage of an AI-powered math app	K-12 STEM education	Students	Individual

Figure 1. (Color online) Conceptual Model



non-DG). Our results show that students living in the epicenter of the outbreak (versus those who do not) initially show a greater reduction in usage of QANDA, a proxy for usage of ICT, but a subsequent increase as time progresses, resembling a J-shaped curve (i.e., compensatory behavior from a quantity perspective). These findings suggest that when first confronted with the pandemic, students expend resources to appraise the disruptive situation and engage in “sense-making” regarding the threat. Such efforts deplete mental resources, making coping a challenge. Therefore, although QANDA usage diminishes initially, students recover over time as they become better able to cope with the new reality and endeavor to return to normalcy. This compensatory behavior for learning loss is consistent with the concept of compensatory consumption behavior, a tendency for consumers to increase the purchase or usage of a product/service with the goal of mitigating self-discrepancy due to a threat (i.e., pandemic crisis) (Mandel et al. 2017).

Moreover, our results provide a more nuanced picture besides quantity by shedding important light on the pattern and pace of compensatory behavior. From a pattern perspective, our results offer a unique understanding into how compensatory behavior for learning loss unfolds. Students living in the epicenter of the outbreak (versus those who do not) display compensatory behavior for learning loss by using the app on a more regular basis. From a pace perspective, findings uncover that students living in the epicenter of the outbreak (versus those who do not) initially fall behind in the math curriculum, but subsequently show compensatory behavior by fully recovering to their normal pace of learning.

In the following sections, we develop our hypotheses. We then explain QANDA, the AI-driven math education app, and the data used to test the hypotheses. Subsequently, we report the results of our hypotheses testing. We conclude with a discussion of research implications along with limitations and directions for future research.

2. Theoretical Background and Hypotheses Development

We develop hypotheses about students’ use of an education app over time in terms of quantity, pattern, and

pace after the pandemic and its variation based on the goal-proximity argument.

2.1. Pandemic Threat

Previous research in illness threat has advanced that outbreaks such as H1N1 “Swine Flu” influenza (Tausczik et al. 2012, Wheaton et al. 2012), Zika virus (Blakey and Abramowitz 2017), severe acute respiratory syndrome (Xie et al. 2011), and Ebola (Blakey et al. 2015, Thompson et al. 2017) are associated with a disruption to people’s mental well-being (e.g., anxiety and stress) and everyday routine life. Students who live in the epicenter of such outbreaks versus those who do not will perceive a smaller psychological distance from the threat, and, hence, the emotional toll is likely to be more acute and far greater when social, educational, and health upheavals occur in the students’ backyard. Therefore, the core of our argument rests on the notion that students will feel more threat from the pandemic when they live in the epicenter of the outbreak (i.e., high pandemic threat) compared with when they do not (i.e., low pandemic threat), and this will have a differential effect on student learning behavior from a tripartite perspective of quantity, pattern, and pace.

As the threat of the pandemic increases for students who live in the epicenter of the outbreak versus those who do not, they will sense a greater loss of control and feel more uncertainty (Smith and Ellsworth 1985). The pandemic will affect the three R’s—routine, ritual, and rule—that define the process of the family resilience framework (Prime et al. 2020). Normal routines (e.g., waiting for the school bus or being dropped off and picked up before and after school) are disrupted, along with other rituals (e.g., family and social gatherings that enrich emotional bonds). Instead, new rules are enforced (e.g., no weekend hangouts and no sleepovers) to help adapt to the new environment. With the threat of the pandemic looming, we advance that students will perceive self-discrepancy—that is, incongruity between how they perceive themselves in that moment and how they desire to view themselves (Higgins 1987). In our context, observing how past routines and taken-for-granted activities are less or no longer permissible, the salience of loss of control and social isolation will be crucial.

However, in order to recover from adversity, students will need to cope with the difficult situation (Lazarus and Folkman 1984). Research has shown that people can cope with adversity by making process-oriented adjustments (Carver 1998). For example, coping has been used in the information systems literature to better understand how users adapt to ICT-induced change from a problem- and emotion-focused perspective (e.g., Beaudry and Pinsonneault 2005, Stein et al. 2015, and Bhattacharjee et al. 2018). Research indicates

that different types of threats (e.g., social rejection, loss of control, or mortality) lead to different coping strategies (i.e., problem-focused or emotion-focused) (Han et al. 2015, Salo et al. 2020). Findings show that loss of control and social rejection trigger the motivation to avoid situations rather than to confront them head-on and lead to more emotion-focused coping strategies (Han et al. 2015, Salo et al. 2020). This is consistent with the Coping Model of User Adaptation, which supports emotion-focused coping when individuals have less control over their situation (Beaudry and Pinsonneault 2005). However, with the passage of time, we posit that students will transition toward problem-focused coping by engaging in compensatory behavior for learning loss.

2.1.1. Compensatory Behavior for Learning Loss.

Coping can be facilitated when two important factors exist: social support (e.g., family and friends, coworkers and peers, and community) and the passage of time (Park 1998). As Rose Fitzgerald Kennedy once said, “Time heals all wounds.” Because students had to consume resources initially to appraise the disruptive situation and engage in emotion-focused coping, time will help replenish the depleted resources, which, in turn, can be used for problem-focused coping. We assert that as time passes, students will begin to engage in problem-focused coping, such as active coping (e.g., “I take direct action to get around the problem”; Carver et al. 1989), planning (e.g., “I make a plan of action”; Carver et al. 1989), seeking instrumental support (e.g., “I try to get advice from someone about what to do”; Carver et al. 1989), and problem solving (e.g., “I come up with a couple different solutions to the problem”; Folkman et al. 1986).

These coping strategies can be summarized as a “direct resolution” type of compensatory behavior (Mandel et al. 2017). That is, although self-discrepancy related to loss of control and social isolation may have weighed more heavily immediately after the outbreak, with the passage of time, students will become increasingly aware of the looming self-discrepancy pertaining to loss of learning. We posit that the motivation to attenuate self-discrepancy from loss of learning will drive compensatory behavior, resulting in students returning to normalcy captured in three different metrics of learning behaviors: (a) the volume of learning, (b) the pattern of learning, and (c) the pace of learning.

First, we argue that the pandemic threat will dissipate over time, with a shift in the saliency of self-discrepancy from that of loss of control and social isolation to that of loss of learning. A natural and common tendency of compensatory behavior is to increase the volume of learning to mitigate self-discrepancy, and we expect this to be more pronounced for students living in the epicenter than

those who do not (Mandel et al. 2017). We expect the usage of the education app to be greater for students who live in the epicenter of the outbreak than for those who do not because there will be a greater sense of urgency to compensate for the greater prior learning loss. Thus, we predict that students living in the epicenter of the outbreak (versus those who do not) will experience a greater initial decrease in the use of an education app followed by a greater rebound in usage with the passage of time (i.e., a J-shaped curve). Therefore, we propose the following compensatory-behavior hypothesis with respect to quantity:

Hypothesis 1 (a). *Students who live in the epicenter of the COVID-19 outbreak (versus those who do not) will initially incur a greater decrease, but subsequently engage in greater compensatory behavior revealed by a greater increase in the use of an education app.*

The second component of the tripartite compensatory behavior is the pattern of compensatory behavior for learning loss. The focus here is on whether students try to make up for their learning loss in a uniform manner (i.e., studying behavior that follows a consistent pattern) or not. We assert that the accessibility of the math solutions provided by an AI-driven education app around the clock (24/7) allows opportunities to be enacted upon and that students who live in the epicenter of the outbreak will engage in a more uniform compensatory-behavior pattern over time than those who do not, based on the following reasons. First, we argue that a uniform studying routine where there is less variability in the amount of time studied over a given time span (e.g., a week) is considered normal and that students who live in the epicenter of the outbreak have deviated from this uniform pattern to a greater degree due to the greater disruption imposed on them. This suggests that for students living in the epicenter (versus those who do not), as normalcy returns over time, the pattern of their compensatory behavior will become more uniform.

Second, engaging in compensatory behavior is not a hedonic experience, such as binge eating or drinking, but one that is problem-focused that requires the exertion of cognitive resources. Greater variability in the allocation of cognitive resources will be considered a departure from the routine that is more taxing to students. Therefore, we argue that students living in the epicenter (versus those who do not) of the outbreak over time will adhere to a “resource pacing” strategy rather than a “resource concentration” strategy because compensatory behavior that follows a pattern of stability and uniformity will be preferred over volatility. Thus, we propose the following compensatory-behavior hypothesis with respect to pattern:

Hypothesis 1 (b). *Students who live in the epicenter of the COVID-19 outbreak (versus those who do not) will initially*

show a nonuniform pattern, but subsequently engage in greater compensatory behavior, demonstrated by a more uniform pattern of learning when using an education app.

The third element of the tripartite compensatory behavior is the rate at which compensatory behavior for learning loss occurs (i.e., pace) in terms of the topics studied along a subject curriculum for students living in the epicenter (versus those who do not) of the outbreak. For example, math is a subject that is made up of some 2,404 topics organized in the curriculum.⁷ The topics range from basic (e.g., fractions) to advanced (e.g., calculus) along the curriculum. Using an education app, students can go backward in the curriculum to review more basic material or move forward to more advanced topics. We state that, immediately following the outbreak, students living in the epicenter will fall behind in the curriculum, but recover over time by advancing forward in the curriculum to catch up to the level of students who do not live in the epicenter of the outbreak.

Advancing forward and making progress along the curriculum demands cognitive resources and effort, both of which are deficient following the outbreak, but especially more so for students living in the epicenter than those who do not. Therefore, with resources depleted to cope with the strain from the outbreak, students living in the epicenter will be left with sparse resources that can be invested to move forward in the curriculum. This suggests that, initially, students living in the epicenter of the outbreak (versus those who do not) will be stagnant or even fall behind in the curriculum. However, as time passes, and as students living in the epicenter regain normalcy, they will engage in compensatory behavior and make up for the lost ground by progressing through the curriculum at a rate equivalent to those who do not live in the outbreak's epicenter. Thus, we propose the following compensatory-behavior hypothesis with respect to pace:

Hypothesis 1 (c). *Students who live in the epicenter of the COVID-19 outbreak (versus those who do not) will initially struggle to move forward in the curriculum, but subsequently engage in greater compensatory behavior, revealed by recovering to their normal pace of learning when using an education app.*

2.2. Goal-Proximity Effect

We define goal-proximity effect as the tendency for students to promptly act upon a goal as the goal becomes more imminent. We posit goal as a moderator that will condition the tripartite aspect of compensatory behavior hypothesized in Hypotheses 1(a)–1(c). In our context, the goal of South Korean students is to perform at the highest level on the centralized

national exam (similar to the SAT and ACT in the United States, though students in South Korea can only take it once when they are in 12th grade). We underscore the importance of goals in ICT research as stated by Majchrzak et al. (2016, p. 272), who argued that “the focus is not on how ICT artifacts can be used, but on the actors’ goals and capabilities related to potential ICT artifact use.” That is, the same ICT artifact can be used in different ways, depending on the goal one holds.

To further shed light on how goals can shape students’ compensatory behavior from a tripartite perspective, we draw from temporal construal theory (Liberman and Trope 1998, Trope and Liberman 2000). According to this theoretical account, when a goal is proximal, students become more prevention-oriented, which urges them to act promptly. As goals draw near, messages that are framed in a prevention-focused (versus promotion-focused) manner carry more weight because the focus shifts from achieving positive outcomes to preventing negative outcomes (Mogilner et al. 2008). For example, with goals approaching, concerns about desirability (e.g., “Why should I strive to accomplish this goal?”) give way to concerns about feasibility (e.g., “How can I go about accomplishing this goal?”) (Mogilner et al. 2008, p. 671). In a product-choice context, when purchase is imminent, prevention-framed products are more attractive than promotion-framed products, and the underlying reason is based on the argument that “when a purchase is proximal, the anticipated pain from failing to fulfill one’s purchasing goal will drive the appeal of prevention-framed over promotion-framed products” (Mogilner et al. 2008, p. 672).

We posit that as the national exam (the goal in this research) draws closer, anticipated concerns about not doing well on the exam will direct a student’s mindset toward a prevention focus. Taking into account the loss of learning that students experienced due to the threat of the pandemic, we argue that the urge to compensate for this loss of learning will be greater for high school (versus middle school) students because high school students will feel greater pressure to prevent further learning loss. The proximity of the goal will prompt high school students, compared with middle school students, to not only make up more for the learning loss, but also to return to a routine studying pattern sooner and advance forward in the subject curriculum at a faster pace. Consequently, we expect a goal-proximity effect from a tripartite compensatory-behavior perspective (i.e., quantity, pattern, and pace) and formally propose the following:

Hypothesis 2. *High school students will recover from learning loss by (a) increasing the use to a greater degree, (b) returning to a uniform learning pattern sooner, and (c)*

advancing forward in the curriculum at a faster pace compared with middle school students when using an education app.

3. Empirical Analysis

3.1. Empirical Context

We test our hypotheses using data from QANDA, a South Korean AI-powered math education app, which has been operating since 2016. Figure 2 provides screenshots of QANDA's search service with an example problem image and its actual search results. A user takes and uploads a picture of a math problem. QANDA then analyzes the captured image using Optical Character Recognition (OCR) and deep-learning technologies to retrieve and present solutions with line-by-line explanations to the matching question free of charge in a matter of seconds. In addition to solvers for simple math formulas, this advanced AI technology enables QANDA to further handle complex math problems that often include heavy text, as well as figures and diagrams. If there are no valid search results, users can ask questions to tutors directly, which is a paid service. We focus on AI-powered search behaviors in this study because the number of questions addressed to tutors is minimal in comparison with the overall amount of search inquiries. As of November 7, 2021, 39 million students around the world have solved more than 2.5 billion math problems using QANDA. Although such remote and AI-powered education technologies can assist and enhance student learning, we note concerns of their unintended use (e.g., academic dishonesty), especially since the pandemic. In Section 3.8, we further discuss this important issue and

illustrate empirical support showing that the majority of students do not associate the app with cheating.

3.2. Data and Variables

Data from this app offer several advantages in studying the impact of the pandemic. First, QANDA is the leading AI-powered education app in South Korea, with over a 60% penetration rate. Based on its popularity among Korean students, QANDA is ideal for studying the impact of COVID-19 on the use of an education app. Second, we infer geographic location from QANDA users' real-time IP addresses, which allows us to determine whether students live in the outbreak's epicenter (i.e., high versus low pandemic threat). We pinpoint the region of an IP address using three reputable services.⁸ Finally, individual student-level data enable us to estimate heterogeneous effects of the pandemic across demographic groups. We investigate how students' responses to the pandemic threat change between high school and middle school students as their proximity to a goal (i.e., the national university entrance exam) impacts their compensatory behavior for learning loss.

We characterize students' compensatory behaviors for learning loss through a tripartite perspective: quantity, pattern, and pace. First, we operationalize quantity by counting the number of AI-powered queries issued by students in QANDA. We aggregate the number of searches over a week at the individual student level. In the main analysis, we implement a log transformation before running an ordinary least squares (OLS) regression due to the skewed nature of search volume.

Second, to measure the pattern of students' usage of QANDA, we compute the Shannon diversity index of the weekly normalized search count vector for student i at week t as $S_{it} = -\sum_{d=1}^7 p_d^{(it)} \log p_d^{(it)}$, where $p_d^{(it)}$ is the number of searches in day d over to that of the entire week ($d = \text{Monday to Sunday}$). Note that $\sum_{d=1}^7 p_d = 1$, and the measure obtains a maximum value when $\mathbf{p} = [p_1 \ p_2 \ \dots \ p_7]$ is uniformly distributed. As the search volume becomes concentrated on fewer days, the measure decreases until it reaches zero when $p_d = 1$ for some d .

Third, to measure the pace in terms of the topics studied along a subject curriculum, we utilize the math-curriculum index that systemically orders topics ranging from basic (e.g., arithmetic) to advanced (e.g., calculus). The index, which is made up of a total of 2,404 math topics that K-12 students learn (i.e., curriculum index ranges from 1 to 2,404), has been developed by the Korea Ministry of Education. Over the years, math education professionals at QANDA have built a large database of question images, for which they have manually labeled their corresponding curriculum index. This database serves as a ground-truth

Figure 2. (Color online) QANDA's Search Service



Notes. (a) Problem image search. (b) Search results.

training data set, and the company uses it to implement a machine-learning model that can reliably tag new questions asked by students with their curriculum index in real-time. Using this functionality, we acquired curriculum-index information for math image queries in our sample. By construction, questions from higher grades and later chapters have higher index values than those from lower grades and earlier chapters. We average out index values over a week for each student. If questions searched by student i in week t have higher index values than those in week $t-1$, we state that she moves forward to advanced topics; otherwise, she goes backward to review basic topics. Appendix A presents examples of topics in the curriculum index.

3.3. Empirical Strategy

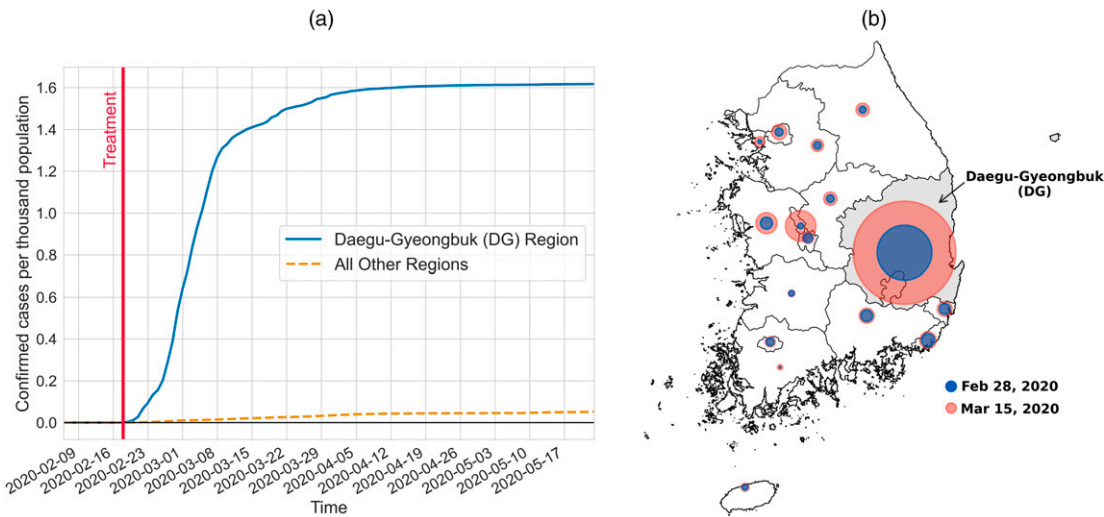
Simple differences in students' learning behavior based on QANDA use before and after the outbreak may be insufficient for estimating the causal effect of COVID-19 because such differences cannot control for any natural changes in students' usage, even in the absence of the shock. For example, during exam periods, student learning activity is high, whereas during breaks, it is low. Aside from seasonality, the platform often sponsored nontargeted promotional events that influenced students' usage, regardless of where they lived. Finally, the overall trend has been an increase in the number of image searches over the years.

To distinguish the causal effect of COVID-19 from possible confounding factors, we exploit exogenous regional variations of the COVID-19 pandemic in

South Korea to create a control group that exhibited substantially lower levels of pandemic threat compared with the treatment group during our sample period. We then compare the differences before and after the outbreak for both the treatment and the control groups, allowing us to net out the time effects (e.g., seasonality and promotional events) across all regions. This strategy relies on the parallel-trend assumption, which requires two groups to follow the same trend before the pandemic. If this assumption holds, we can establish the counterfactual trajectory (i.e., trend of the control group) for the treatment after the pandemic, and the deviation from the trajectory can be attributed to the causal effect of COVID-19.

We now describe how we define the control and treatment groups and the timing of the treatment. South Korea maintained control of the spread of COVID-19 until February 18, 2020. However, its 31st confirmed patient participated in a mass religious gathering at "Shincheonji Church of Jesus" in Daegu, a metropolitan city in the Gyeongbuk province.⁹ Many of the patient's close contacts became infected as well, triggering a drastic escalation of COVID-19 cases in the DG region, whereas the number of cases in all other regions remained relatively stable (see Figure 3). Specifically, the number of cases in DG was zero before February 18 and increased to 338 by February 22 and to 2,724 by February 29. Meanwhile, the number of cases in all other regions only increased to 95 by February 22 and to 426 by February 29. Thus, there is no other epicenter in South Korea except DG. Using economic statistics and Google Trends, we

Figure 3. (Color online) The Spread of COVID-19 in South Korea



Source. Korea Disease Control and Prevention Agency, Statistics Korea.

Notes. (a) Confirmed cases over time. (b) Confirmed cases across administrative regions. Panel (a) shows the trend of cumulative confirmed cases per thousand population for the DG region and all other regions in South Korea over time. Panel (b) shows confirmed cases on February 28, 2020, and March 15, 2020, across administrative regions, where the DG region is shaded in gray, and circle size corresponds to the cumulative confirmed cases per thousand population.

demonstrate in Appendix B that contamination of the control group, if any, is not a major issue.

The spread of the COVID-19 pandemic in the DG region can be considered exogenous. Most of the infections in the DG region are linked to services of the Shincheonji. It is estimated to have approximately 20,000 members across Korea, and about 1,300 members are registered in the DG region. The number of members per capita is lower in the DG region than in other regions (Aum et al. 2021). Thus, the regional variation in pandemic threat (i.e., infections) in South Korea is free of any confounding factors that may affect both infections and students' use of QANDA. This natural experiment setting allows us to employ a difference-in-differences (DID) model, where students in DG are the treatment group and students in all other regions are the control group, with the starting point of the treatment being the fourth week of February (February 16, 2020–February 22, 2020).

Our sample period spans from the third week of December (December 15, 2019–December 21, 2019) to the fourth week of May (May 17, 2020–May 23, 2020). We estimate ordinary least squares on a random sample of 36,000 active users who made at least five searches using QANDA during the pretreatment period. Table 2 provides the descriptive statistics of the three variables of interest (i.e., quantity, pattern, and pace) for our sample by region (DG versus non-DG), grade (all, middle, and high school), and stage. To examine model-free evidence, we partitioned our data into three stages by time: Stage 0 includes weeks before the COVID-19 outbreak in DG (i.e., the

prepandemic period before the week of February 16, 2020); stage 1 corresponds to the period in which we observe learning loss from the week of February 16 to the week of March 29; and stage 2 contains the remaining weeks of the learning-recovery period. Overall, there is an increasing trend in all three variables over time. However, from stage 0 to stage 1, the DG region conveys a smaller increase than the non-DG region for all three variables, indicating that the DG region had a slower growth immediately after the COVID-19 outbreak. In stage 2, we can see that the variables for the DG region recover to similar or higher levels compared with the levels of the non-DG region as a sign of compensating for the slow growth in the previous stage. Similar patterns can also be observed when viewing by grade as shown in Table 2, but are more pronounced for high school students. In Appendix C, we also provide summary statistics of the entire data set for all variables, including the correlation matrix.

3.4. Econometric Specification

First, we implemented a standard DID (i.e., without leads and lags) to estimate the net (total) effect of the pandemic threat on learning behavior. We estimate the following equation:

$$y_{irt} = \gamma_i + \lambda_t + \beta \text{Treat}_i \times \text{Post}_t + \mathbf{X}'_{rt} \boldsymbol{\delta} + \varepsilon_{irt},$$

where the subscripts i , r , and t denote student, region, and week, respectively. y_{irt} denotes the dependent variable: log number of searches (i.e., quantity), Shannon diversity index of weekly search counts (i.e., pattern), and curriculum index (i.e., pace). γ_i denotes

Table 2. Descriptive Statistics of the Three Variables of Interest by Region, Stage, and Grade.

Variable	Grade	DG region			Non-DG region		
		Stage 0 (prepandemic)	Stage 1 (learning loss)	Stage 2 (learning recovery)	Stage 0 (prepandemic)	Stage 1 (learning loss)	Stage 2 (learning recovery)
Quantity (log of search volume)	All	7.09 (17.57)	8.30 (21.71)	13.57 (25.42)	7.50 (18.86)	10.19 (24.29)	13.28 (25.66)
	High	5.71 (16.08)	6.96 (19.56)	13.09 (24.26)	5.71 (16.32)	8.43 (21.47)	12.04 (23.57)
	Middle	8.62 (18.96)	9.77 (23.77)	14.11 (26.62)	9.53 (21.20)	12.20 (27.00)	14.69 (27.79)
Pattern (search diversity index)	All	0.24 (0.46)	0.29 (0.51)	0.49 (0.58)	0.25 (0.47)	0.32 (0.53)	0.46 (0.57)
	High	0.20 (0.43)	0.26 (0.49)	0.51 (0.59)	0.21 (0.44)	0.29 (0.51)	0.46 (0.57)
	Middle	0.29 (0.49)	0.32 (0.52)	0.48 (0.58)	0.30 (0.50)	0.36 (0.54)	0.45 (0.57)
Pace (curriculum index)	All	1,474.37 (436.80)	1,513.30 (451.53)	1,577.39 (445.76)	1,489.39 (438.87)	1,538.55 (438.68)	1,575.91 (428.41)
	High	1,822.48 (331.25)	1,840.93 (348.31)	1,875.76 (316.35)	1,831.16 (320.63)	1,853.00 (318.51)	1,862.62 (298.45)
	Middle	1,197.36 (286.61)	1,208.15 (296.11)	1,232.54 (301.45)	1,202.69 (296.22)	1,231.87 (301.17)	1,253.56 (305.63)

Notes. Numbers represent means, and standard deviations are in parentheses. Level of analysis is individual-week.

individual fixed effects, and λ_t denotes time fixed effects. $Treat_i$ is the treatment variable that equals one if the student i lives in DG and zero otherwise, and $Post_t$ is a dummy variable that takes the value of one if the week t is after the fourth week of February (February 16, 2020–February 22, 2020). X_{rt} is a vector of dynamic region-specific controls that includes weather information such as weekly average temperature, the number of days with rain per week, and the number of days with snow per week. ε_{irt} is the idiosyncratic shock to student i in region r at week t . The DID coefficient β captures the net effect of the pandemic threat.

The above specification does not capture how the pandemic threat affects the use of an AI-powered learning app over time. For our main hypotheses testing, we used a DID model with leads-and-lags to investigate the dynamic effects and examine the parallel-trend assumption in the pretreatment period. Specifically, we estimate the following equation using weekly individual-level data:

$$y_{irt} = \gamma_i + \lambda_t + \sum_{j=-9}^{13} \beta_j D_{jrt} + X'_{rt} \delta + \varepsilon_{irt},$$

where D_{jrt} is a vector of dummy variables that equals one if the observation falls j weeks before or after the fourth week of February and region r is DG, and zero otherwise. The key coefficient of interest is the vector of β_j , designed to capture the effect of COVID-19. All other variables are the same as the previous specification for the net effect. We estimate the regression coefficients for the whole sample and also separately by student grade, specifically, high versus middle school students.

We note that elementary school students are excluded from the heterogeneous treatment-effect analysis, as they can be more susceptible to sampling bias than middle school and high school students,

resulting in biased parameter estimates. In accordance with the premise that voluntary participation is the cornerstone of ethical research on human subjects (Patel et al. 2003), users in our study downloaded and used the AI-powered learning app of interest voluntarily. Nonetheless, such voluntary participation invariably generates self-selection bias (Holmes 2004). Even though there is no required informed-consent process by which a fully informed and competent individual voluntarily decides whether to download and use the app in our setting, we expect significant parental involvement in the selection and use of children's apps, including the focal learning app. This perspective is mainly compatible with the literature on child computer interaction (e.g., Papadakis et al. 2019), in which researchers reveal that parental background and own experiences are important factors that can influence children's engagement with digital media. Because younger children are less competent to make informed decisions regarding the use of digital learning tools (Patel et al. 2003), parental involvement is more likely to be prevalent among younger age groups, such as elementary school students, than among older age groups, and, thus, goal proximity is less relevant for younger users. In light of this, we expect that the sampling bias resulting from elementary school students' involuntary participation will be greater than that of other age groups; hence, elementary school students were excluded from the heterogeneous treatment-effect analysis.

3.5. Results

Table 3 presents the net effect on each dependent variable: log number of searches (i.e., quantity), Shannon diversity index of weekly search counts (i.e., pattern), and curriculum index (i.e., pace). When search volume is used as the dependent variable, the net effect is close to zero and statistically insignificant ($p > 0.1$). For pattern, the net effect is positive, suggesting that

Table 3. The Net Effect by DID Estimation

Variables	Quantity			Pattern		
	All	High	Middle	All	High	Middle
<i>Treat × Post</i>	−0.002 (0.018)	0.017 (0.024)	−0.018 (0.028)	0.013** (0.006)	0.022** (0.009)	0.009 (0.010)
<i>Average temperature</i>	−0.015*** (0.003)	−0.016*** (0.004)	−0.015*** (0.004)	−0.007*** (0.001)	−0.007*** (0.001)	−0.008*** (0.002)
<i>Number of days with rain</i>	−0.009*** (0.002)	−0.010*** (0.002)	−0.008*** (0.003)	−0.004*** (0.001)	−0.005*** (0.001)	−0.004*** (0.001)
<i>Number of days with snow</i>	0.021*** (0.004)	0.037*** (0.005)	0.014** (0.007)	0.010*** (0.002)	0.014*** (0.002)	0.009*** (0.002)
Observations	828,000	403,719	356,201	828,000	403,719	356,201

Notes. Robust standard errors are in parentheses. The estimator is OLS with individual and time fixed effects. The dependent variables are the log number of searches (i.e., quantity) and the diversity index of weekly searches (i.e., pattern). We omit the estimation results of the net effect for pace because the curriculum index is a cumulative measure.

** $p < 0.05$; *** $p < 0.01$.

the pandemic threat resulted in a more uniform and equally distributed studying pattern of QANDA use. However, this effect is significant ($p < 0.05$) only for high school students. For pace, it is worth noting that the curriculum index is, by definition, a cumulative measure. That is, as long as the curriculum index returns to its original course at some point during our sample period, we can conclude that the overall effect is zero. Hence, it is not appropriate to estimate the net effect to determine whether the pandemic threat leads to overall learning loss (or gain) in terms of pace, as was the case for search volume and pattern. Rather, we focus on the dynamic effect for pace.

Although the net effect results are interesting, they mask any dynamic changes in students' learning over time. For instance, based on the fact that the net effect on quantity is insignificant, one cannot determine whether (a) the pandemic threat did not affect students' learning at all or (b) learning loss was canceled out by learning gain during the sample period. Figure 4 presents the estimation results of the dynamic effects for all samples in terms of quantity, pattern, and pace to uncover changes in students' learning behavior due to the pandemic threat. We find that our parallel-trend assumption holds, and our quasi-experiment setting is well-designed, as shown by a flat trend in the pretreatment period.¹⁰ We use the 95% confidence interval for interpreting the results throughout the paper.

Hypothesis 1(a) reasons that students near the epicenter of the outbreak will show less use of an education app initially after the COVID-19 outbreak, but more as time progresses. The regression coefficients in the first column of Table 3 confirm this hypothesis. We observe less search volume using QANDA in the first five weeks (0 ~ +4) following the COVID-19 outbreak in the DG region. Over time, the negative effect dissipates, and positive effects begin in the ninth week (+9) after the outbreak. Specifically, the number of searches declines by approximately 10.2%, on average, followed by an increase of 12.7%, on average. As Figure 4(a) shows, the dynamic effects in the post-treatment period resemble a J-shaped curve. These results support students' compensatory behavior in terms of quantity, as stated in Hypothesis 1(a).

Hypothesis 1(b) posits that students near the epicenter of the outbreak will initially show a nonuniform studying routine, but a more uniform pattern, as normalcy returns over time. Thus, the diversity index of weekly searches will decrease, but subsequently increase as search volumes are evenly distributed over each day of the week. As Figure 4(b) depicts, the diversity index decreases in week 0, but subsequently returns to the prepandemic level. That is, students closer to the outbreak deviate from the uniform pattern of study in the week of the outbreak due to the

disruption imposed on them. However, the effect dissipates immediately, resulting in no effect a week after the outbreak. Students begin to engage in a more uniform studying pattern after the eighth week (+8) of the outbreak, compared with the prepandemic level, as shown in Figure 4(b). These results support students' compensatory behavior in terms of pattern, as stated in Hypothesis 1(b).

Hypothesis 1(c) states that students near the outbreak's epicenter will first fall behind in the subject curriculum, but will subsequently recover to their normal pace of learning. As shown in Figure 4(c), the curriculum index decreases a week after the outbreak, but catches up to the prepandemic level in the seventh week (+7) after the outbreak. This suggests that, in the aftermath of the outbreak, students near the epicenter initially regress in the curriculum, but, with time, rebound to a curriculum path equivalent to students who do not live in the outbreak's epicenter. These results support students' compensatory behavior in terms of pace, as stated in Hypothesis 1(c).

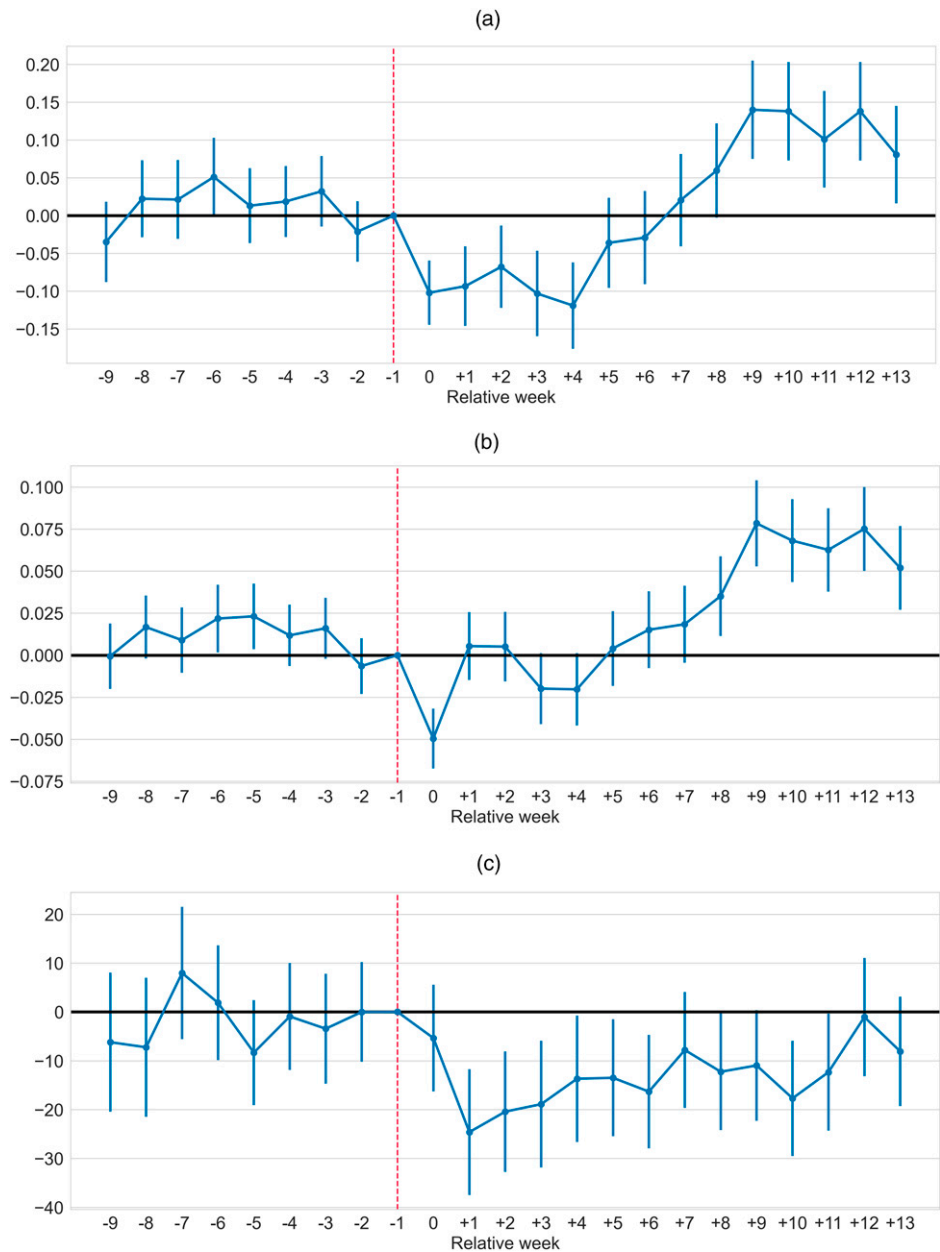
Hypothesis 2 predicts that high school students will recover from learning loss by (a) increasing the use to a greater degree (quantity), (b) following a more uniform studying pattern (pattern), and (c) advancing forward in the curriculum at a faster rate (pace) compared with middle school students. When goals (e.g., the national university entrance exam) are near (versus far), students act upon the opportunity more promptly, leading to a greater recovery from learning loss.

Figure 5(a) illustrates that high school students (versus middle school students) demonstrate greater compensation for learning loss in terms of quantity. Whereas high school students return to normalcy in the fifth week (+5) following the outbreak, middle school students return to normalcy in the seventh week (+7). After they return to normalcy, the number of searches for high school students jumped by 18.4%, whereas it increased by just 5.8% for middle school students.

In terms of pattern, high school students study in a more uniform and evenly distributed manner. As shown in Figure 5(b), the diversity index decreased in the week of the outbreak, but immediately returned to the prepandemic level for both high and middle school students. However, high school students show a uniform studying pattern sooner and to a greater extent as they start to compensate for their learning loss. The diversity index for high school students rose above its prepandemic level in the seventh week (+7) after the outbreak, whereas it surpassed its prepandemic level for middle school students in the ninth week (+9).

In terms of pace, the curriculum index for high school students initially declined, but recovered to the normal curriculum course by the third week (+3) following the outbreak; however, the curriculum index for middle school students did not recover to a normal

Figure 4. (Color online) Dynamic Effects of the Pandemic Threat by DID Estimation with Leads-and-Lags



Notes. (a) Quantity. (b) Pattern. (c) Pace. Vertical bars denote 95% confidence intervals. The dependent variables are quantity (log number of searches) (a), pattern (diversity index of weekly searches) (b), and pace (curriculum index) (c). The estimator is OLS with individual and time fixed effects. The dummy variable indicating a week before treatment in DG is omitted as the base group and denoted as a dashed vertical line. The estimation result tables are provided in Appendix D.

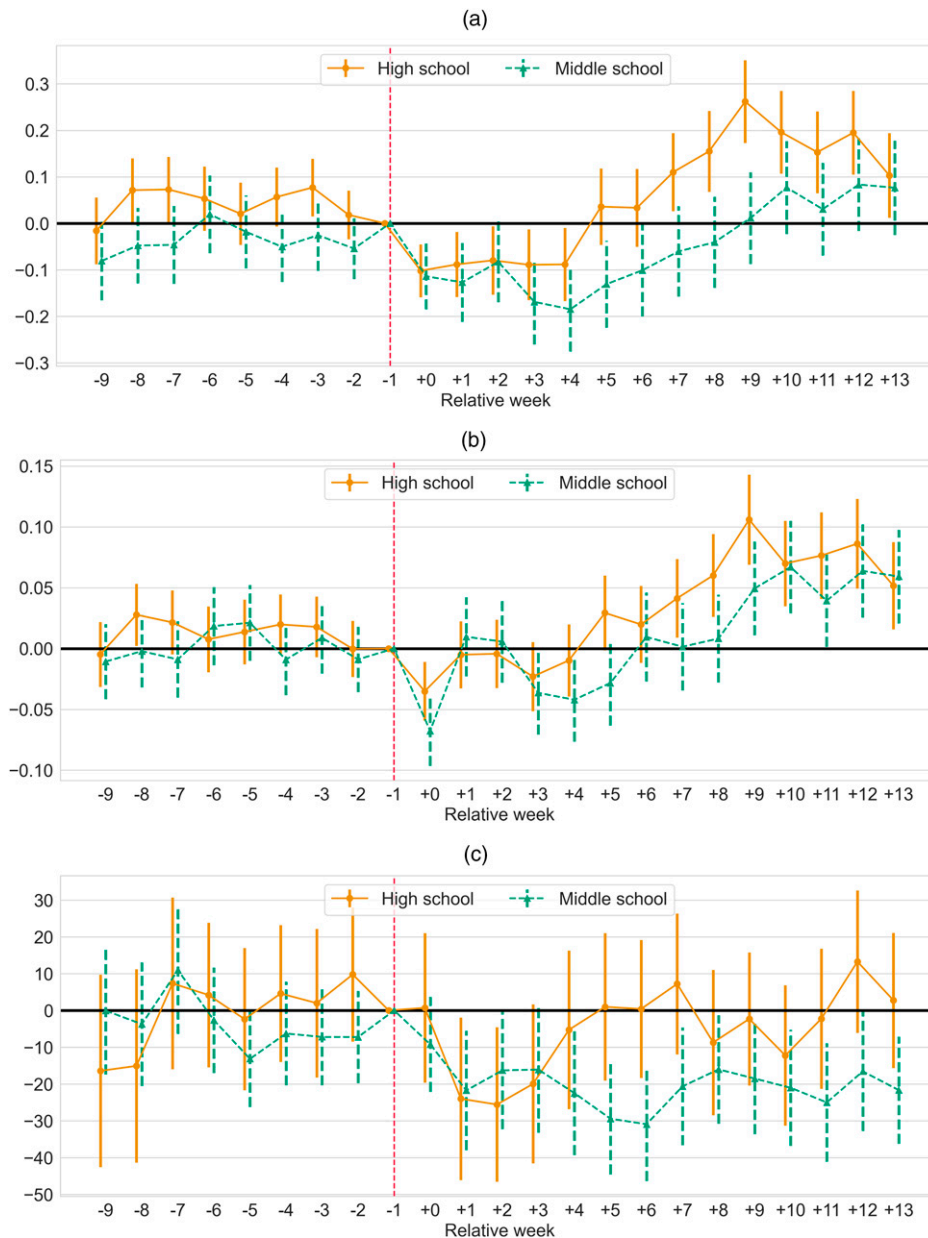
path during our sample period. Figure 5 in tabular format is provided in Appendix D.

To formally test for the equality of coefficients between the two samples (i.e., high and middle school students) in the posttreatment period, we conducted z-tests. The z-scores for posttreatment periods are reported in Table 4. The equality of coefficients is only rejected within particular time periods (i.e., five periods

in terms of quantity, three periods in terms of pattern, and five periods in terms of pace at the 0.05 significance level). Thus, these findings lend partial support to Hypothesis 2.

3.6. Alternative Explanations

One could argue that changes in QANDA use are attributable to school-closure policies, rather than to

Figure 5. (Color online) Dynamic Effects of the Goal Proximity

Notes. (a) Quantity. (b) Pattern. (c) Pace. Vertical bars denote 95% confidence intervals. The dependent variables are the log number of searches (a), the diversity index of weekly searches (b), and the curriculum index (c). The estimator is OLS with individual and time fixed effects. The dummy variable indicating a week before treatment in DG is omitted as the base group and denoted as a dashed vertical line. The estimation result tables are provided in Appendix D.

students' perceived pandemic threat. It is worth noting that the varying degree of perceived pandemic threat between DG and non-DG regions is not confounded with school-closure rules, as all K-12 schools in Korea were forced to close on March 2, 2020, per government order, and the schools were on winter break from the last week of December 2019 to March 1, 2020. Further, school openings cannot also be considered a confounding factor to the causal relationship. Remote learning began in schools in April 2020,

and it was a nationwide policy with no regional variations. As a result, there was no actual change in school closures and openings over the course of our study period (December 15, 2019–May 23, 2020). Lastly, schools reopened in-person classes on a staggered schedule beginning the last week of May 2020, resulting in geographical variations. To account for this, we exclude samples collected after the last week of May 2020.

Another possible explanation is that the closure of cram schools may affect the use of QANDA, which

Table 4. Testing for Equality of Coefficients Between High and Middle School Students in the Posttreatment Period

Variable	Quantity	Pattern	Pace
0 weeks since COVID-19 outbreak	0.25	1.71*	0.81
1 weeks since COVID-19 outbreak	0.69	−0.68	−0.16
2 weeks since COVID-19 outbreak	0.05	−0.45	−0.69
3 weeks since COVID-19 outbreak	1.31	0.59	−0.28
4 weeks since COVID-19 outbreak	1.57	1.40	1.24
5 weeks since COVID-19 outbreak	2.63***	2.42**	2.37**
6 weeks since COVID-19 outbreak	2.04**	0.42	2.53**
7 weeks since COVID-19 outbreak	2.62***	1.62	2.19**
8 weeks since COVID-19 outbreak	2.93***	2.05**	0.58
9 weeks since COVID-19 outbreak	3.69***	2.11**	1.35
10 weeks since COVID-19 outbreak	1.74*	0.11	0.70
11 weeks since COVID-19 outbreak	1.81*	1.41	1.79*
12 weeks since COVID-19 outbreak	1.62	0.82	2.32**
13 weeks since COVID-19 outbreak	0.39	−0.28	2.04**

Note. The estimator is z-score: $z = (\beta_{\text{high}} - \beta_{\text{mid}}) / \sqrt{\text{Var}(\beta_{\text{high}}) + \text{Var}(\beta_{\text{mid}})}$.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

are private institutions that train students to achieve particular goals such as performance in entrance exams to universities. About 95% of cram schools in the DG region closed between mid and late February, whereas about 30% of cram schools in Seoul, the capital city of Korea, closed during the same period. On the one hand, if QANDA is a substitute for cram schools, search volumes on the app would have risen in the DG region following the COVID-19 outbreak because students were unable to attend cram schools. Given that the use of QANDA fell (rather than rose) in the DG region (versus the non-DG region) in the first five weeks following the outbreak, we believe that this explanation is implausible. On the other hand, if QANDA is used as a complement to cram schools (i.e., students use QANDA to help them with their cram-school homework), then its use would have decreased following the outbreak. Although this is consistent with our finding that QANDA search volumes decreased in the first five weeks of the outbreak, it is not consistent with the increase in QANDA use in the ninth week and subsequent weeks because the closure rate of cram schools in the DG region remained relatively high, at 37%.¹¹ Thus, we can rule this out as an alternative explanation. To further alleviate concerns that cram schools may have confounded the use of QANDA, we gathered cram-school closure rates, included them as additional control variables, and performed the main DID with leads-and-lags analysis. When cram-school closure rates are included as controls, as shown in Appendix E (see Table E.2), the results remain largely consistent and robust.¹²

3.7. Robustness Checks

Our results are robust to a variety of alternative specifications. First, because our dependent variable for quantity is a count variable, we replicate estimations using

Poisson regression, rather than assuming a logged OLS functional form. The Poisson estimation yields results that are consistent with the main specification.

Second, we conduct additional analyses using time-stamped data.¹³ Specifically, we use the average (per problem) and total time spent for reviewing problem solutions as alternative dependent variables. The results demonstrate that high school students show a greater compensatory behavior compared with middle school students during the recovery period, in terms of time spent in learning (see Figures E.1 and E.2. in Appendix E). These results indicate that high school students indeed put in more effort in multiple dimensions, thereby providing additional support for the goal-proximity effect.

Third, we replicate our estimation for quantity after excluding outliers with more than the 99th percentile (1,679) of searches using QANDA. There are no significant changes in estimates, confirming that our findings are not influenced by outliers.

Fourth, unless there is at least one search in a week, there are no observations on the curriculum index. To account for this, we employ a Heckman's two-step model (Heckman 1979) to capture the possible non-random selection mechanism. In the first stage, we run a probit regression with the dependent variable equaling one if the curriculum index is not missing.¹⁴ In the second stage, we run the same regression as the main specification, but include the inverse Mills ratio (IMR) from the first stage as an additional control variable. As shown in Appendix E (see Table E.1), the coefficient of IMR is negative and statistically significant ($p < 0.01$), indicating that the selection correction term adjusts the curriculum index downward. Nonetheless, the resulting bias for the coefficients of interest seems minimal, as the coefficient estimate of pace is consistent with that of our main specification, where selection bias is not corrected for.

Fifth, we consider additional time-varying region-specific control variables. Two significant economic factors are added to our model specification—namely, employment and retail sales, both of which are measured monthly by region and available from Statistics Korea.¹⁵ As shown in Appendix E (see Table E.3), the results remain qualitatively consistent.¹⁶

Lastly, we conducted a subsample analysis, focusing on students who have used other learning services of the app (e.g., quiz questions, asking a tutor, concept books, etc.), along with the main search service. They make up about 43.6% of students in our data set. Results are consistent and robust for all three variables of interest, as can be seen in Appendix E (see Table E.5).

3.8. Alternative Purposes of Using the App

One might be concerned whether students are primarily using QANDA for cheating purposes. To address

this concern, we performed a multifaceted investigation that includes (i) additional analyses of QANDA usage data and (ii) a confidential student survey conducted by a third-party reputable survey company.¹⁷

First, we analyzed how students use the app's search functionality at three different stages in time: presearch, intrasearch (during), and postsearch. Figure 6 highlights the measurements that reflect different aspects of how students use the search functionality at each stage.

For the presearch stage, we examined problem images that students uploaded to the app to see whether they contained any handwritten notes, as a sign of students' attempt to solve the problem on their own before searching in QANDA. We manually labeled 3,000 randomly selected problem images from August 2021 and found that more than 50% of them contained some handwritten notes. This shows that the majority of students attempted to solve the problem themselves (i.e., scribbled) before approaching QANDA.

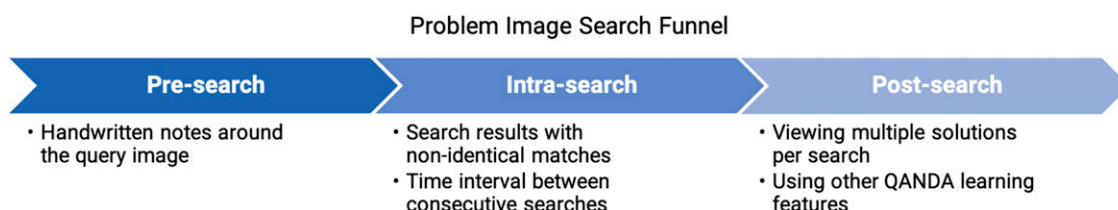
For the intrasearch stage, we investigated the search results that were shown to the students. Using deep-learning technologies, QANDA provides multiple search results by identifying problems in its database that match students' query images and presenting line-by-line solutions to the matching problem in the order of similarity scores (from the largest to smallest). A large similarity score between the query image and the matching problem may imply that they are identical, which would immediately give students the exact answer to their query. However, it could also be the case that there are minor variations between the two such as different variable names, numbers, or mathematical operators (e.g., see the example search in Figure 2, where the query image and matched problem are highly similar, but not identical). In the latter case, students would need to understand and apply the search result to their original query in order to derive the correct answer. Against this backdrop, we randomly selected more than 21,000 query images and manually compared each query image to the matching problem in the database. Upon such a comparison, we found that just 16.8% of them were exact, identical matches, mitigating the likelihood that students use the app solely for cheating purposes. Separately, we examined QANDA's user-activity-log data for our sample period. When just copying responses

from search results, patterns of quick consecutive searches are likely to emerge. As a result, we calculated the time interval between successive searches for each user. On average, 85.0% of users have a time delay of more than two minutes between consecutive searches, indicating that they are less likely to rapidly copy the solution and continue searching for their next query. When increasing the time delay to more than three minutes, we still observe a similar pattern, with 76.3% of users having a delay of more than three minutes between consecutive searches. Overall, these results suggest that the majority of students exhibit QANDA usage patterns that are less associated with cheating. Additionally, we conducted subsample analyses for students who had a time interval between consecutive searches longer than two and three minutes (see Figures F.1 and F.2 in Appendix F, respectively). The results are consistent and robust with the main analysis in terms of quantity, pattern, and pace.

For the postsearch stage, we further probed QANDA's observational data. QANDA gives multiple search results ranked by the aforementioned similarity score (i.e., the first search result has the largest similarity score to the query image). Thus, if a student is only looking for quick solutions, he has less motivation to look at other possible solutions beyond the first search result. However, we found that more than 50% of users view multiple solutions per search, rather than just the top search result. Separately, we discovered that about 43.6% of students use the app's other learning features (e.g., quiz questions, asking a tutor, concept books, etc.) in addition to the core image-based search functionality, reducing the possibility that students are using QANDA solely for the purpose of cheating.

Second, we conducted a confidential online survey asking students how they or their peers use QANDA, which was commissioned through a third-party reputable survey company to ensure that the survey would be objective and unbiased. We asked a stratified random sample of 400 students, which matches the gender and grade composition of students in South Korea, to answer the following three questions on a five-point scale: (i) Q1: "How helpful is QANDA when studying alone?" (1: very unhelpful, 5: very

Figure 6. (Color online) Measurements at Different Stages of the Problem Image Search Funnel



helpful); (ii) Q2: “How much do you pay attention to the line-by-line explanations in the search results?” (1: not at all, 5: very much); and (iii) Q3: “How much do your peers use QANDA for copying solutions?” (1: not at all, 5: very much). Following prior research, we used indirect questioning for the last survey question, which asks about unethical behavior, to mitigate the effects of social desirability bias (Fisher 1993, King and Bruner 2000). The response rate was 55.5%, and we did not find any statistical differences between the response and nonresponse groups using two-sample *z*-tests in terms of their demographics: gender ($p = 0.494$), grade ($p = 0.908$), and region ($p = 0.642$).

Survey results (i.e., top two box scores, responses 4 or 5, on a 1–5 scale) show that 71.5% of students indicated that QANDA is helpful or very helpful, and 81.5% answered that they carefully examine the line-by-line explanations of the search results. For the last survey question, less than one out of four students (24.3%) reported that their peers use QANDA for just copying solutions. As a robustness check, we compared early versus late responses (Armstrong and Overton 1977) and did not find any statistical differences in the responses when splitting the respondents into two equally sized groups based on how soon they responded to the survey for all survey questions (Q1: $p = 0.888$, Q2: $p = 0.525$, and Q3: $p = 0.696$). Hence, the survey results suggest that most students do not associate the app with cheating purposes.

4. General Discussion

The COVID-19 global pandemic has brought about immense changes in our lives. Among the many areas it has challenged, the education sector has been particularly vulnerable. In particular, K-12 students have been profoundly affected in terms of how they need to adjust their learning behaviors. Although there has been growing interest in how the crisis has affected student learning, several important gaps have yet to be studied. This research attempts to fill some of these gaps.

This paper sheds new light on the following two questions that motivated us to pursue this line of research: (1) How do students compensate for learning loss during a pandemic? (2) What is the role of AI in this regard? We also asked a follow-up question that was more fine-grained: Is there a difference in compensatory behavior for learning loss when using an AI-powered learning app in terms of quantity, pattern, and pace of usage depending on (a) the level of pandemic threat and (b) the proximity of a goal to students? The current paper demonstrates that pandemic threat influences students’ compensatory behaviors for learning loss through the use of an AI-powered learning app. In this section, we discuss the theoretical

implications of our empirical findings and managerial implications for the EdTech sector.

Lockdowns may not have hurt the economy as much as the perceived threat from the pandemic (Goolsbee and Syverson 2021). Consider the cases of South Korea and Taiwan, two countries that never imposed a formal government lockdown, but managed the crisis through the scrupulous use of the three T’s (testing, tracing, and treatment) to mitigate the threat felt by the pandemic. Notwithstanding, our data and our findings indicate within-country variation (in terms of how close students are living to the hotspot) in terms of the heterogeneous effect of the pandemic threat on student learning behavior. Our data show that the pandemic outbreak disproportionately disrupts ontological security for students who live closer to the outbreak by disrupting students’ sense of order and meaning and their social framework (Campbell et al. 2020). Students close to the outbreak experience uprooted norms and routines that normally function as a “firewall against chaos” (Giddens 1991, p. 40). However, we also find that these students show exceptional resilience by engaging in greater compensatory behavior for learning loss, compared with students farther away from the outbreak. Based on the results of this research, we now discuss how our findings contribute to the compensatory-behavior literature.

4.1. Theoretical Contribution to Compensatory-Behavior Literature

Our research contributes to the compensatory-behavior literature. Although extant literature on compensatory behavior has taken a static view by focusing on the alignment between a single self-discrepancy (e.g., loss of control or social rejection) and a coping strategy (i.e., emotion- or problem-focused) (Mandel et al. 2017), our paper takes a dynamic perspective to elucidate how some self-discrepancies may be transient, allowing for coping strategies to change over time (Salo et al. 2020). Our research adds to the literature on information technology (IT) users’ coping strategy that has examined how users navigate between emotion-focused and problem-focused coping strategies after negative IT incidents (Salo et al. 2020). We took a tripartite view of compensatory behavior by examining (a) search volume to gauge “how much” students used the education app (i.e., quantity), (b) “how” students engaged in compensatory behavior (i.e., pattern), and (c) “the rate” at which they navigated the subject curriculum (i.e., pace). Such a 360-degree approach illuminates a holistic understanding of students’ road to normalcy that prior research has not been able to offer.

Our finding that loss of learning was greater for students in the DG (versus the non-DG) region soon after the outbreak suggests that the threat of the pandemic initially led to loss of control and social isolation self-discrepancies. In other words, the emphasis was on self-

discrepancies that are fundamental to who students are as human beings, rather than self-discrepancy to learning actors caused by loss of learning. Thus, the coping strategy employed to mitigate such self-discrepancies was more emotion-focused. However, with the passage of time, as students began to adjust to the “new normal,” our findings show that the loss-of-learning self-discrepancy became more salient and that students could proactively address this by engaging in more use of an AI-powered education app. Indeed, students did adopt a more problem-solving coping approach using direct resolution (e.g., increase in the use of QANDA through more search volume) to compensate for the learning loss.

This study also uncovered the pattern of compensatory behavior, an aspect of compensatory behavior that had yet to be examined in the literature. Most empirical research in compensatory behavior has relied on experiments, rather than longitudinal data. Therefore, it was not possible to observe or test how compensatory behavior unfolded over time. The focus was on a “snapshot” at a particular point in time, limiting how the pattern of compensatory behavior changed over time. To address such shortcomings, our data allowed us to monitor and view over a 13-week period the pattern of how students engaged in compensatory behavior on their road to recovery. We find that the pattern of compensatory behavior of students closer to the epicenter (versus those farther away) returned to a more uniform and regular studying pattern over time. These patterns reflect students’ desire to return to stability after the outbreak, especially for students who were harder hit. We believe that the unique characteristic (i.e., 24-7 accessibility) of the opportunity offered by the AI-powered education app played an instrumental role in the enactment process by allowing students to study more regularly. Because students had unrestricted access to the line-by-line solutions and were not pressed to use the service within a narrow window of opportunity, students were able to evenly distribute their cognitive resources in a manner that minimized the strain, given their limited resource capacity.

Finally, our data speak to the pace that students navigated to compensate for learning loss. By developing a curriculum index of math that systematically orders topics ranging from basic to advanced, we were able to discern what content students focused on immediately after the outbreak versus later during the path to normalcy. Results suggest that students who felt a higher level of pandemic threat lagged those who perceived a lower level of pandemic threat soon after the outbreak by falling behind in the curriculum, but compensated for their learning loss by advancing forward in the curriculum over time to eventually catch up. These findings are in line with the compensatory-

behavior literature, which shows that what is perceived as a self-discrepancy (e.g., physical appearance, intelligence, or power) is what is sought after (Rucker and Galinsky 2008, Hoegg et al. 2014, Kim and Gal 2014). In our study, it was moving forward in the math curriculum by studying topics further along in the curriculum to recover to their normal pace of learning. A point of departure of our study from the rest of the literature is that increased pandemic threat accelerated this process, whereas goal proximity to a national exam partially precipitated this process.

A final word on the moderating role of goal proximity is in order, as it received partial support. Quantity, in terms of search volume, received partial support as goal proximity played a moderating role, evidenced by high school students engaging in more search volume than middle school students, but not throughout the entire period. Similarly, pattern and pace also received only partial support, as high school students did not show a faster return to a uniform learning pattern or a quicker progression along the curriculum index compared with middle school students throughout the 13-week period. Although it is difficult to definitively discern why pattern and pace did not vary based on goal proximity, we do acknowledge that the goal of this study was an external goal that the student themselves did not set, but was, rather, imposed on them. Our results could have been better supported had the goal setting been volitional and set by themselves, as goal-setting research suggests that goal setting is a motivational process (Locke and Latham 1990).

4.2. Managerial Implications

From a practical perspective, our research offers implications for (1) teachers, parents, and students; and (2) EdTech firms. First, our findings suggest that an AI-based education app, such as QANDA, can be especially beneficial to students who desire to voluntarily recover from adversity when they perceive a high level of crisis and when they have a goal that is imminent. Parents and teachers may encourage students who fall into these situations to use education apps to complement their learning outside the classroom.

However, there are also potential risks of using such education software. From an ethical viewpoint, the possibility of misusing apps (i.e., for cheating purposes) to finish homework cannot be ignored. Students may become blindly dependent on apps for schoolwork, reducing the motivation to work hard and weakening their work ethic to solve problems on their own. Although these are legitimate concerns, based on our thorough investigation, the evidence points to directions against cheating when using QANDA. Results from analyzing observational data (presearch, intrasearch, and postsearch) and the user survey collectively suggest that cheating is not the main purpose for using QANDA.

These results are reassuring for parents and teachers, who desire to safeguard students from unintended use of the app, and to EdTech firms that wish to grow their business. These findings corroborate education apps as complementary learning tools outside of the classroom that can help students supplement in-school learning, especially more so for students whose lives have been disrupted due to adversity.

The findings about the goal-proximity effect provide practical implications for platform design. The notion of nudging in behavioral economics can be effectively applied to leverage the power of goals. Known as the “progress illusion” effect, the mere perception of progress facilitates goal attainment. For example, in a classic study (Kivetz et al. 2006), coffee shop customers were given one of two loyalty stamp cards, offering a free cup of coffee after the 10th cup. Card A had 10 empty slots for stamps, whereas Card B had 12 slots for stamps but two were already pre-stamped. Results showed that customers with Card B earned a free cup of coffee faster than those who had Card A.

The above results suggest that simply monitoring and tracking progress toward a goal can affect behavioral change. Recording food intake can curb food consumption and facilitate weight loss. Fitness wearables that track and monitor health, such as Fitbit and Kore, use progress-tracking bars and weekly reports to alert the status of where users are on their path to goal attainment. We recommend that EdTech firms can similarly provide progress reports and tracking information, where the starting point is not at 0%, but, rather, at 20% to further motivate students to use the app in ways that will help them achieve goals. Also, for students that are falling behind in their studies due to ailing health, financial distress, or family instability, EdTech firms can send push notifications outlining near and attainable goals that students can actualize, contributing to faster learning-loss recovery.

Furthermore, app developers can design apps so that students can easily establish and include proximal and distal goals into their daily study routine. In developing new products, if developers desire to experiment or test a new feature and would like to receive feedback on its functionality, incentivizing students with a near or within-reach attainable goal (versus a far, abstract, and unrealistic goal) can be an effective way to collect feedback relatively quickly.

Another implication of our research from a public policy perspective is the total effect of COVID-19 on student learning. We find that students fully compensated from learning loss over time when using QANDA. In terms of quantity (search volume), students made up for the initial decrease in search volume following the outbreak by revealing an increase in search volume with time, to the point where the

total effect was zero. This suggests that the front-loaded learning loss was compensated by back-loaded learning gain. For pattern, the total effect was positive, implying that students developed a more uniform and consistent studying habit over time since the pandemic. Regarding pace, students were able to progress along the curriculum index to prepandemic level, despite falling behind initially following the outbreak, suggesting that they made up ground on the math topics they should be learning. These results are encouraging, as they collectively point to policy makers and EdTech managers that AI-powered education apps can play a pivotal role in mitigating learning loss and help students get back on the road to normalcy.

Recently, the Biden Administration announced the Build Back Better (BBB) framework and signed the \$1.2 trillion Infrastructure Bill into law.¹⁸ In the \$1.2 trillion bill, \$65 billion is allocated to expanding and upgrading the broadband infrastructure to rural parts of the country, where access to the internet is challenging. Also, one of the key components of the BBB package is to expand access to affordable, high-quality education beyond high school. We believe that with more widespread and affordable access to broadband, AI-powered education apps such as QANDA can benefit many students and families that are struggling with providing quality and affordable education, especially during these trying times.

Furthermore, the World Bank estimates that this generation of students is expected to lose an estimated \$10 trillion in earnings, equivalent to 10% of global GDP, as a consequence of increased learning loss and student dropouts (World Bank 2021). As part of the World Bank’s continuing effort to assist countries in recovering from learning loss due to COVID-19, it is pursuing a three-phase model: coping (while schools are closed), manage continuity (as schools reopen), and improve and accelerate (throughout, seize opportunities to improve system for the long term) (World Bank 2020). There is ample opportunity for AI-powered learning apps to contribute to such endeavors, as we have come to realize the importance and value of building and leveraging multimodal technology, such as AI and digital platforms. To this end, AI-powered apps in EdTech will increasingly play a pivotal role that shapes how quality and affordable education can be effectively, equitably, and resiliently delivered to students, parents, teachers, and the broader education community.

4.3. Limitations and Future Research Directions

This research is not without limitations, which provide directions for future research. First, when testing the goal-proximity effect, we assumed that the goal students had in mind was the national university entrance exam, which is held every year on the third Thursday of November in South Korea. This was an

external goal imposed on all students. Our research did not address goals that were imposed internally or self-set, such as achieving 90% or higher on a midterm or final exam. The national university exam goal served the purpose of this research well because it is a common goal shared across all students in South Korea, with proximity (depending on grade level) being the only difference. Future studies could examine whether and how our results might change when internal academic achievement goals are set by the students themselves.

Second, the importance of the goal may have played a factor in our research. In South Korea, the national university entrance exam is the main determining factor to which university students get accepted. However, in the United States, SAT or ACT test scores, albeit important, play a less major role in the decision-making process for granting admissions. Therefore, if a goal is proximal, but perceived to be of less significance, or there is heterogeneity across individuals on the perceived importance of goals, whether the goal-proximity effect will still be observed is an open question that awaits future research.

Third, although we theorized and empirically showed that the temporal proximity to goals drove the results for the goal-proximity effect, it is difficult to rule out the possibility that age may have played a role. That is, high school students, being older than middle school students, are more mature. Older, more mature students may be able to recover from distress quicker and, accordingly, engage in greater compensatory behavior for learning loss sooner. However, age is highly correlated with years of schooling (here, grade level) in adolescence in cultures where formal schooling is pervasive (Cahan and Cohen 1989, Bisanz et al. 1995). In South Korea, it is mandatory for children between the ages of six and 15 to attend school. Although high school is not compulsory, 99.7% of teenagers between 15 and 17 years of age attend high school, according to the Korea Ministry of Education. For this reason, it is impractical to separate the effect of age from the effect of age-related variables in this study. We acknowledge this as a limitation of our data, potentially confounding the effect of goals, and excluding us from ruling out age as an alternative explanation. Nonetheless, we note as supporting circumstantial evidence that prior studies point to a much stronger grade effect than age effect on intellectual and academic performances (Cahan and Cohen 1989, Cliffordson and Gustafsson 2008, Luyten et al. 2008, Cliffordson 2010, Mavilidi et al. 2022). Moreover, Nam (2014) suggests in a study on Korean students that age effect tends to be attenuated, and younger students focus more on their academic studies, especially in secondary education, due to the fierce academic competition for university entry. Thus, we encourage future

research to tease apart age and grade level to more clearly determine which factor is at work in explaining the goal-proximity effect.

Fourth, although the primary focus of this research was not to examine whether cheating is the primary motivation for using QANDA, we did make a concerted effort to further explore this possibility by conducting a thorough and detailed analysis of additional observational data and administering a user survey to rule out the tendency that students may use QANDA for cheating purposes. Notwithstanding lacking evidence to support cheating, the long-term effects of how students use the app are unclear and warrant future research. We believe that the long-term effect of how students use such an app (e.g., cheating) should be examined with ground-truth data in a more systematic and rigorous manner.

Fifth, although the focus of our research was on an AI-powered math learning app, the potential for a learning app to assist students in other subject areas should be explored. For example, other STEM subjects in science could be candidates, where some of the topics covered could also involve math (e.g., physics and chemistry). Furthermore, because the OCR feature used by this AI-powered math learning app was able to address quite lengthy word problems, this capability opens up the possibility to address a diverse range of problems in various subjects expressed in different modalities.

Lastly, this study mainly relied on observational QANDA usage data, which has limitations in examining the causal impact of QANDA usage on students' task performance (i.e., math grades) and nontask performance (i.e., enjoyment and fear of math learning), due to potential self-selection bias. As a result, future investigations based on randomized controlled experiments will be required to establish the causal impact of QANDA usage on students' task and nontask performances.

5. Conclusion

The current study finds that students who use an AI-powered education app are resilient in the face of adversity. More crucially, we demonstrate *how much*, *how regularly*, and *how quickly* learning recovery unfolds over time, depending on the severity of pandemic threat and how close (versus far) a goal is. The findings from this research contribute to the IS literature by improving our understanding of compensatory-learning behavior when using an AI-powered math education app. We hope that this research sparks more interest and follow-up research on how students' use of education technologies changes in the face of adversity.

Acknowledgements

The authors thank the excellent review panel, which was led by D. J. Wu and included an anonymous associate editor and

three reviewers. The authors are also grateful to Xueming Luo, Min-Seok Pang, Yang Wang, Kunsoo Han, Shun Ye, Jingyuan Yang, Antino Kim, Daegon Cho, Abhishek Ray, Min Chen, Eun Ju Jung, Sabari Karmegam, Heeseung Yang, Eunchoel Shin, Jae-Hwuen Jung, Sunghun Chung, Elina Hwang, Elia Rho, Jiyong Park, and On-Ook Oh for helpful comments and feedback on earlier versions of the manuscript. The authors also thank seminar participants at Temple University, KAIST, Korea University, and George Mason University for their valuable comments. The authors also gratefully acknowledge Ray Lee and Jake Lee of Mathpresso, Inc., for their support in providing the data for this study. S. P. Han acknowledges that he serves as an advisor to Mathpresso, Inc., and thanks Ben Yoo, Hyun Lee, Koo Chung, Paul Yu, Ella Shim, and Gyu Lee of

Mathpresso, Inc., for their research assistance. The usual disclaimer applies.

Appendix A. Examples of Curriculum Index

The math-curriculum index systemically orders topics ranging from basic (e.g., arithmetic) to advanced (e.g., calculus). The index, which is made up of a total of 2,404 math topics that K-12 students learn (i.e., curriculum index ranges from 1 to 2,404), has been developed by the Korea Ministry of Education. By construction, questions from higher grades and later chapters have higher index values than those from lower grades and earlier chapters. Table A.1 provides examples of the curriculum index and its corresponding math topic.

Table A.1. Examples of the Curriculum Index

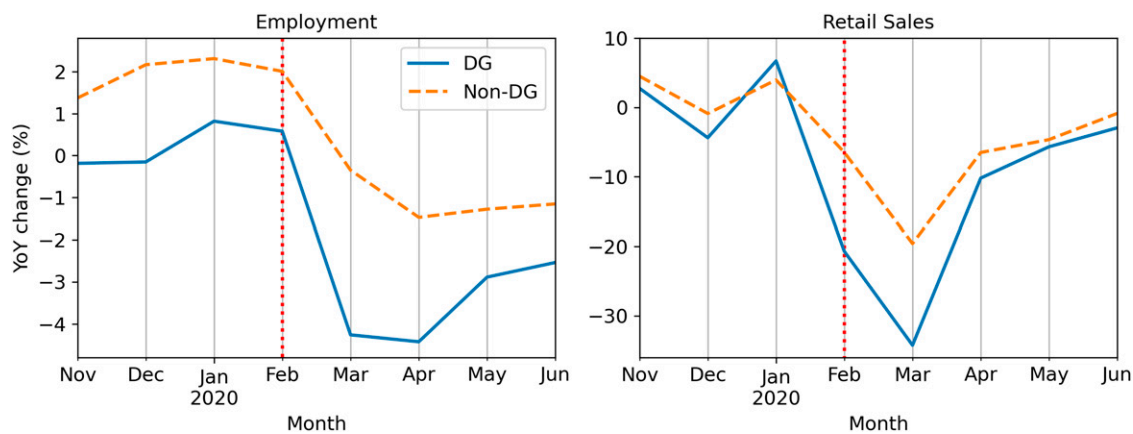
School	Group	Topic	Curriculum index
Elementary	Number and Operations	Five-digit numbers	3
	Geometry	Isosceles triangle	135
	Number and Operations	Addition/subtraction of fractions	251
	Geometry	Hexahedron	392
	Measurement	Volume of cube	522
Middle	Geometry	Pythagorean theorem	650
	Expressions and Equations	Linear equation	770
	Geometry	Circular cylinder	930
	Expressions and Equations	Simultaneous equation	1054
	Geometry	Parallelogram	1157
	Functions	Quadratic function	1371
	Geometry	Angle of circumference	1500
	Expressions and Equations	Quadratic inequality	1629
High	Limit of Sequences	Convergence/divergence of sequence	1850
	Integration	Integration by parts	1974
	Exponent and Logarithm	Quadratic equation and logarithm	2100
	Calculus	Power rule	2303

Appendix B. Different Levels of Pandemic Threat in the Control and Treatment Groups

We illustrate the different levels of pandemic threat in the control (i.e., non-DG region) and treatment (i.e., DG region) groups. Particularly, we show changes in (1) monthly employment and retail sales from Statistics Korea, as well as (2) weekly search keyword trends from Google Trends¹⁹ for the DG and non-DG regions. In all cases, we use year-over-year (YoY) change percentages to eliminate seasonality and obtain a clearer trend of the time series. In Figure B.1, we can see that both employment and retail sales have a much

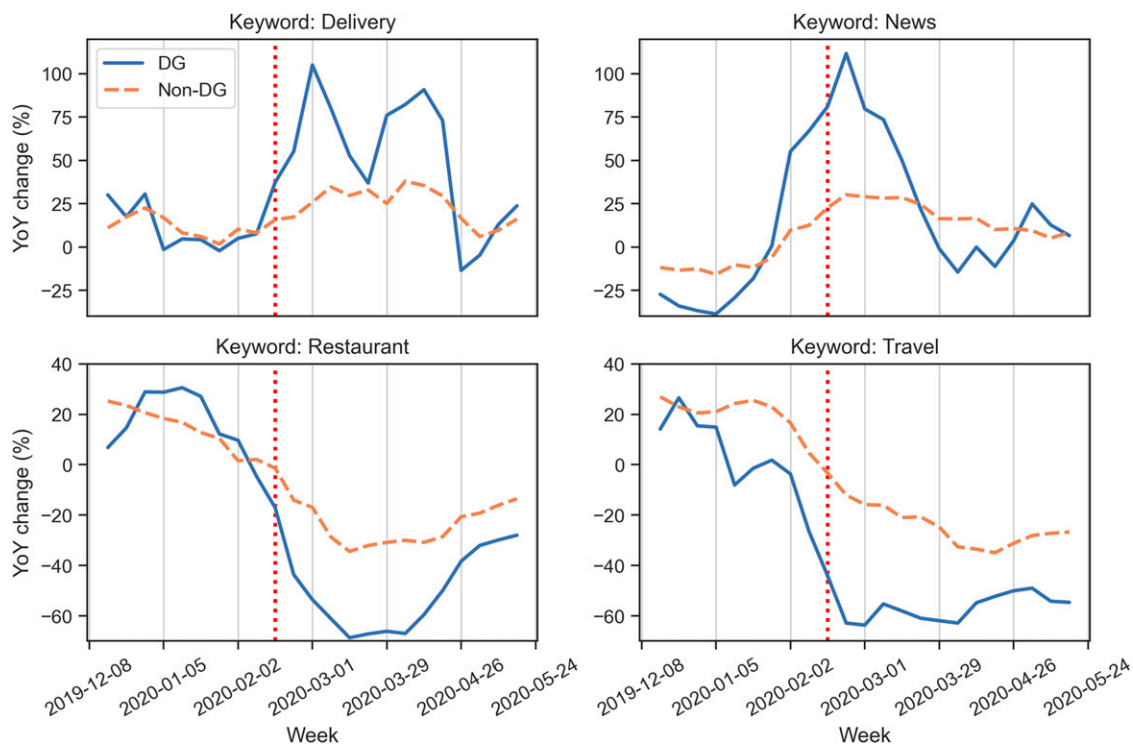
larger drop in the DG region than in the non-DG region immediately following the initial COVID-19 outbreak. For the search keyword trends, we selected keywords that are highly related to the pandemic, including “Delivery,” “News,” “Restaurant,” and “Travel.” In Figure B.2, we can see that the search keywords “Delivery” and “News” show a significant increase, whereas “Restaurant” and “Travel” show a substantial decrease in the DG region compared with the non-DG region. These figures collectively demonstrate that contamination of the control group, if any, is not a major issue.

Figure B.1. (Color online) Monthly Employment and Retail Sales for DG and Non-DG Regions



Note. The treatment month in DG is denoted as a dotted vertical line.

Figure B.2. (Color Online) Weekly Google Trends for DG and Non-DG Regions of Pandemic-Related Search Keywords (Delivery, News, Restaurant, and Travel)



Note. The treatment week in DG is denoted as a dotted vertical line.

Appendix C. Summary Statistics

Table C.1 presents summary statistics of the entire data set for all variables, including the correlation matrix.

Table C.1. Summary Statistics

Variable	Obs.	Mean	Std. dev.	Correlation						
				1	2	3	4	5	6	7
1. Search count	828,000	9.96	22.72	1.00						
2. Search diversity index	828,000	0.33	0.53	0.44	1.00					
3. Curriculum index	322,943	1,501.41	453.58	−0.08	−0.01	1.00				
4. DG	828,000	0.09	0.29	−0.01	0.00	−0.01	1.00			
5. Average temperature	828,000	7.52	5.75	0.05	0.10	0.07	0.07	1.00		
6. Number of days with rain	828,000	2.36	1.40	−0.01	0.00	0.01	−0.05	0.13	1.00	
7. Number of days with snow	828,000	0.13	0.50	0.00	0.00	0.00	−0.08	−0.35	0.22	1.00

Notes. Level of analysis is individual-week. The number of observations (Obs.) pertaining to the curriculum index is smaller than that associated with the other variables. The primary reason for missing values in the curriculum index is that there are no searches over a week. To alleviate the potential selection bias, we estimate a Heckman selection model for robustness check and find consistent results as in the main analysis. Details of the Heckman selection results are provided in Section 3.7. Std. dev., standard deviation.

Appendix D. The Dynamic Impact of the Pandemic Threat on the Use of QANDA

Table D.1 contains results shown in Figure 4, and Tables D.2, D.3, and D.4 contain results shown in Figure 5 in tabular format.

Table D.1. The Dynamic Impact of the Pandemic Threat on QANDA Use (All Samples)

Dependent variable	Quantity	Pattern	Pace
9 weeks until COVID-19 outbreak	−0.0348 (0.0270)	−0.000651 (0.00991)	−6.194 (7.259)
8 weeks until COVID-19 outbreak	0.0223 (0.0259)	0.0167* (0.00953)	−7.240 (7.265)
7 weeks until COVID-19 outbreak	0.0213 (0.0266)	0.00895 (0.00991)	7.965 (6.915)
6 weeks until COVID-19 outbreak	0.0511* (0.0264)	0.0218** (0.0102)	1.880 (5.992)
5 weeks until COVID-19 outbreak	0.0131 (0.0252)	0.0231** (0.00995)	−8.331 (5.486)
4 weeks until COVID-19 outbreak	0.0187 (0.0240)	0.0118 (0.00933)	−0.928 (5.579)
3 weeks until COVID-19 outbreak	0.0322 (0.0238)	0.0160* (0.00927)	−3.439 (5.734)
2 weeks until COVID-19 outbreak	−0.0211 (0.0204)	−0.00647 (0.00847)	−0.0159 (5.209)
1 week until COVID-19 outbreak		Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.102*** (0.0219)	−0.0496*** (0.00909)	−5.361 (5.570)
1 week since COVID-19 outbreak	−0.0933*** (0.0268)	0.00541 (0.0103)	−24.61*** (6.576)
2 weeks since COVID-19 outbreak	−0.0677** (0.0278)	0.00507 (0.0106)	−20.42*** (6.294)
3 weeks since COVID-19 outbreak	−0.103*** (0.0289)	−0.0199* (0.0108)	−18.89*** (6.619)
4 weeks since COVID-19 outbreak	−0.119*** (0.0292)	−0.0203* (0.0109)	−13.68** (6.599)
5 weeks since COVID-19 outbreak	−0.0360 (0.0303)	0.00400 (0.0113)	−13.49** (6.102)
6 weeks since COVID-19 outbreak	−0.0291 (0.0314)	0.0151 (0.0116)	−16.32*** (5.927)
7 weeks since COVID-19 outbreak	0.0205 (0.0312)	0.0184 (0.0117)	−7.799 (6.060)
8 weeks since COVID-19 outbreak	0.0597* (0.0320)	0.0351*** (0.0121)	−12.25** (6.089)
9 weeks since COVID-19 outbreak	0.140*** (0.0328)	0.0784*** (0.0129)	−10.96* (5.781)
10 weeks since COVID-19 outbreak	0.138*** (0.0328)	0.0681*** (0.0126)	−17.69*** (6.023)
11 weeks since COVID-19 outbreak	0.101*** (0.0328)	0.0626*** (0.0127)	−12.33** (6.101)
12 weeks since COVID-19 outbreak	0.138*** (0.0332)	0.0751*** (0.0129)	−1.051 (6.171)

Table D.1. (Continued)

Dependent variable	Quantity	Pattern	Pace
13 weeks since COVID-19 outbreak	0.0805** (0.0331)	0.0519*** (0.0127)	−8.079 (5.705)
Region controls: weather	Yes	Yes	Yes
Observations	828,000	828,000	322,943

Notes. Robust standard errors are in parentheses. The dependent variables are the log number of searches (i.e., quantity), (b) the diversity index of weekly searches (i.e., pattern), and (c) the curriculum index (i.e., pace). The estimator is OLS with individual and time fixed effects. The number of observations pertaining to the curriculum index as a dependent variable is smaller than that associated with the other dependent variables. The primary reason for missing values in the curriculum index is that there are no searches over a week. To alleviate the potential selection bias, we estimate a Heckman selection model for robustness check and find consistent results as in the main analysis. Details of the Heckman selection results are provided in Section 3.7.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table D.2. The Impact of the Pandemic Threat on Quantity using QANDA by Grade

Subsample	High	Middle
9 weeks until COVID-19 outbreak	−0.0160 (0.0366)	−0.0810* (0.0430)
8 weeks until COVID-19 outbreak	0.0713** (0.0349)	−0.0478 (0.0415)
7 weeks until COVID-19 outbreak	0.0728** (0.0358)	−0.0463 (0.0426)
6 weeks until COVID-19 outbreak	0.0531 (0.0354)	0.0194 (0.0424)
5 weeks until COVID-19 outbreak	0.0206 (0.0342)	−0.0182 (0.0401)
4 weeks until COVID-19 outbreak	0.0566* (0.0324)	−0.0505 (0.0386)
3 weeks until COVID-19 outbreak	0.0771** (0.0315)	−0.0257 (0.0390)
2 weeks until COVID-19 outbreak	0.0181 (0.0267)	−0.0543 (0.0334)
1 week until COVID-19 outbreak		Omitted (baseline)
0 weeks since COVID-19 outbreak	−0.102*** (0.0289)	−0.114*** (0.0358)
1 week since COVID-19 outbreak	−0.0884** (0.0358)	−0.127*** (0.0433)
2 weeks since COVID-19 outbreak	−0.0797** (0.0376)	−0.0829* (0.0444)
3 weeks since COVID-19 outbreak	−0.0892** (0.0388)	−0.169*** (0.0466)
4 weeks since COVID-19 outbreak	−0.0884** (0.0401)	−0.185*** (0.0463)
5 weeks since COVID-19 outbreak	0.0358 (0.0419)	−0.131*** (0.0477)
6 weeks since COVID-19 outbreak	0.0334 (0.0425)	−0.101** (0.0505)
7 weeks since COVID-19 outbreak	0.110*** (0.0424)	−0.0604 (0.0496)
8 weeks since COVID-19 outbreak	0.155*** (0.0444)	−0.0408 (0.0501)
9 weeks since COVID-19 outbreak	0.262*** (0.0457)	0.0114 (0.0503)
10 weeks since COVID-19 outbreak	0.196*** (0.0455)	0.0769 (0.0512)
11 weeks since COVID-19 outbreak	0.153*** (0.0450)	0.0304 (0.0508)
12 weeks since COVID-19 outbreak	0.195*** (0.0460)	0.0833 (0.0511)
13 weeks since COVID-19 outbreak	0.103** (0.0460)	0.0766 (0.0519)
Region controls: weather	Yes	Yes
Observations	403,719	356,201

Notes. Robust standard errors are in parentheses. The dependent variable is the log number of searches (i.e., quantity). The estimator is OLS with individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table D.3. The Effect of COVID-19 on Pattern using QANDA by Grade

Subsample	High	Middle
9 weeks until COVID-19 outbreak	−0.00481 (0.0136)	−0.0107 (0.0157)
8 weeks until COVID-19 outbreak	0.0278** (0.0130)	−0.00215 (0.0152)
7 weeks until COVID-19 outbreak	0.0215 (0.0134)	−0.00894 (0.0160)
6 weeks until COVID-19 outbreak	0.00761 (0.0138)	0.0185 (0.0164)
5 weeks until COVID-19 outbreak	0.0137 (0.0135)	0.0212 (0.0159)
4 weeks until COVID-19 outbreak	0.0198 (0.0126)	−0.00910 (0.0149)
3 weeks until COVID-19 outbreak	0.0178 (0.0127)	0.00880 (0.0149)
2 weeks until COVID-19 outbreak	−0.000238 (0.0117)	−0.00891 (0.0137)
1 week until COVID-19 outbreak	Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.0351*** (0.0123)	−0.0677*** (0.0147)
1 week since COVID-19 outbreak	−0.00506 (0.0140)	0.00968 (0.0166)
2 weeks since COVID-19 outbreak	−0.00428 (0.0143)	0.00570 (0.0172)
3 weeks since COVID-19 outbreak	−0.0230 (0.0145)	−0.0364** (0.0174)
4 weeks since COVID-19 outbreak	−0.00973 (0.0151)	−0.0421** (0.0175)
5 weeks since COVID-19 outbreak	0.0293* (0.0157)	−0.0283 (0.0179)
6 weeks since COVID-19 outbreak	0.0198 (0.0161)	0.00951 (0.0187)
7 weeks since COVID-19 outbreak	0.0413** (0.0164)	0.00148 (0.0183)
8 weeks since COVID-19 outbreak	0.0601*** (0.0174)	0.00823 (0.0184)
9 weeks since COVID-19 outbreak	0.106*** (0.0186)	0.0494** (0.0197)
10 weeks since COVID-19 outbreak	0.0699*** (0.0180)	0.0670*** (0.0193)
11 weeks since COVID-19 outbreak	0.0764*** (0.0179)	0.0393** (0.0193)
12 weeks since COVID-19 outbreak	0.0862*** (0.0187)	0.0638*** (0.0196)
13 weeks since COVID-19 outbreak	0.0517*** (0.0183)	0.0592*** (0.0197)
Region controls: weather	Yes	Yes
Observations	403,719	356,201

Notes. Robust standard errors are in parentheses. The dependent variable is the diversity index of weekly searches (i.e., pattern). The estimator is OLS with individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table D.4. The Effect of COVID-19 on Pace Using QANDA by Grade

Subsample	High	Middle
9 weeks until COVID-19 outbreak	−16.42 (13.35)	−0.0745 (8.822)
8 weeks until COVID-19 outbreak	−15.07 (13.40)	−3.720 (8.554)
7 weeks until COVID-19 outbreak	7.360 (11.92)	10.95 (8.824)
6 weeks until COVID-19 outbreak	4.196 (10.03)	−2.659 (7.298)
5 weeks until COVID-19 outbreak	−2.360 (9.863)	−13.15** (6.677)
4 weeks until COVID-19 outbreak	4.602 (9.483)	−6.257 (7.182)
3 weeks until COVID-19 outbreak	1.998 (10.31)	−7.208 (6.681)
2 weeks until COVID-19 outbreak	9.779 (9.301)	−7.216 (6.396)
1 week until COVID-19 outbreak	Omitted (baseline)	
0 weeks since COVID-19 outbreak	0.713 (10.36)	−9.212 (6.592)
1 week since COVID-19 outbreak	−24.00** (11.26)	−21.74*** (8.281)
2 weeks since COVID-19 outbreak	−25.57** (10.71)	−16.30** (8.147)
3 weeks since COVID-19 outbreak	−19.96* (11.02)	−16.06* (8.757)
4 weeks since COVID-19 outbreak	−5.248 (10.98)	−22.47*** (8.572)
5 weeks since COVID-19 outbreak	0.984 (10.22)	−29.42*** (7.716)
6 weeks since COVID-19 outbreak	0.407 (9.566)	−30.93*** (7.864)
7 weeks since COVID-19 outbreak	7.225 (9.758)	−20.62** (8.155)
8 weeks since COVID-19 outbreak	−8.703 (10.07)	−16.02** (7.532)
9 weeks since COVID-19 outbreak	−2.298 (9.226)	−18.51** (7.702)
10 weeks since COVID-19 outbreak	−12.24 (9.727)	−21.02*** (8.043)
11 weeks since COVID-19 outbreak	−2.235 (9.703)	−25.00*** (8.197)
12 weeks since COVID-19 outbreak	13.28 (9.873)	−16.56** (8.274)
13 weeks since COVID-19 outbreak	2.745 (9.365)	−21.66*** (7.430)
Region controls: weather	Yes	Yes
Observations	150,731	154,179

Notes. Robust standard errors are in parentheses. The dependent variable is the curriculum index (i.e., pace). The estimator is OLS with individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Appendix E. Robustness Checks

Table E.1. Alternative Specifications

Dependent variable	Model		
	Poisson Quantity	Without outliers Quantity	Heckman Pace
9 weeks until COVID-19 outbreak	−0.0827 (0.0542)	−0.0382 (0.0269)	—
8 weeks until COVID-19 outbreak	0.0406 (0.0504)	0.0180 (0.0259)	−7.347 (7.300)
7 weeks until COVID-19 outbreak	0.0631 (0.0507)	0.0172 (0.0266)	6.801 (6.903)
6 weeks until COVID-19 outbreak	0.0696 (0.0426)	0.0503* (0.0264)	0.831 (5.995)
5 weeks until COVID-19 outbreak	0.00897 (0.0406)	0.0148 (0.0252)	−9.217* (5.489)
4 weeks until COVID-19 outbreak	0.0381 (0.0406)	0.0168 (0.0240)	−1.830 (5.567)
3 weeks until COVID-19 outbreak	0.0603 (0.0393)	0.0287 (0.0238)	−5.109 (5.732)
2 weeks until COVID-19 outbreak	−0.0356 (0.0341)	−0.0208 (0.0204)	−0.349 (5.200)
1 week until COVID-19 outbreak		Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.193*** (0.0382)	−0.101*** (0.0219)	−6.111 (5.564)
1 week since COVID-19 outbreak	−0.107* (0.0587)	−0.0946*** (0.0269)	−24.88*** (6.575)
2 weeks since COVID-19 outbreak	−0.0657 (0.0557)	−0.0675** (0.0278)	−20.52*** (6.295)
3 weeks since COVID-19 outbreak	−0.111** (0.0534)	−0.104*** (0.0290)	−19.16*** (6.619)
4 weeks since COVID-19 outbreak	−0.134** (0.0526)	−0.119*** (0.0292)	−13.84** (6.584)
5 weeks since COVID-19 outbreak	−0.0776 (0.0480)	−0.0370 (0.0304)	−13.61** (6.102)
6 weeks since COVID-19 outbreak	−0.0137 (0.0487)	−0.0299 (0.0315)	−16.74*** (5.919)
7 weeks since COVID-19 outbreak	0.00971 (0.0479)	0.0211 (0.0312)	−8.551 (6.058)
8 weeks since COVID-19 outbreak	0.0352 (0.0451)	0.0571* (0.0321)	−13.25** (6.098)
9 weeks since COVID-19 outbreak	0.105** (0.0441)	0.140*** (0.0329)	−11.23* (5.775)
10 weeks since COVID-19 outbreak	0.143*** (0.0449)	0.134*** (0.0329)	−18.39*** (6.035)
11 weeks since COVID-19 outbreak	0.0945** (0.0430)	0.0925*** (0.0329)	−13.55** (6.098)
12 weeks since COVID-19 outbreak	0.135*** (0.0426)	0.134*** (0.0333)	−1.212 (6.172)
13 weeks since COVID-19 outbreak	0.0864** (0.0435)	0.0777** (0.0332)	−8.669 (5.705)
Inverse Mills ratio			−7.73*** (0.924)
Region controls: weather	Yes	Yes	Yes
Observations	828,000	819,766	313,577

Notes. Robust standard errors are in parentheses. In the Heckman model, the dummy variable indicating nine weeks before treatment in DG is omitted due to the inclusion of the lag variable of the number of searches. All specifications include individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table E.2. Results with Additional Dynamic Region-Specific Controls: Cram School Closure Rates

Dependent variable	Quantity	Pattern	Pace
9 weeks until COVID-19 outbreak	−0.0347 (0.0270)	−0.000615 (0.00991)	−4.650 (7.259)
8 weeks until COVID-19 outbreak	0.0199 (0.0260)	0.0154 (0.00957)	−5.677 (7.276)
7 weeks until COVID-19 outbreak	0.0180 (0.0267)	0.00720 (0.00993)	8.711 (6.923)
6 weeks until COVID-19 outbreak	0.0474* (0.0264)	0.0200* (0.0102)	1.636 (5.994)
5 weeks until COVID-19 outbreak	0.0114 (0.0256)	0.0225** (0.0101)	−7.956 (5.516)
4 weeks until COVID-19 outbreak	0.0139 (0.0240)	0.00964 (0.00937)	−1.832 (5.587)
3 weeks until COVID-19 outbreak	0.0213 (0.0240)	0.0104 (0.00939)	−5.390 (5.814)
2 weeks until COVID-19 outbreak	−0.0242 (0.0204)	−0.00727 (0.00849)	0.266 (5.208)
1 week until COVID-19 outbreak		Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.112*** (0.0221)	−0.0534*** (0.00917)	−6.433 (5.596)
1 week since COVID-19 outbreak	−0.0749** (0.0303)	0.00180 (0.0116)	−20.29*** (7.109)
2 weeks since COVID-19 outbreak	−0.0371 (0.0317)	0.00581 (0.0120)	−14.72** (6.917)
3 weeks since COVID-19 outbreak	−0.0635** (0.0323)	−0.0167 (0.0120)	−13.39* (7.148)
4 weeks since COVID-19 outbreak	−0.0727** (0.0327)	−0.0108 (0.0122)	−5.218 (7.210)
5 weeks since COVID-19 outbreak	−0.0138 (0.0362)	−0.00117 (0.0134)	−6.607 (7.178)
6 weeks since COVID-19 outbreak	−0.00962 (0.0362)	0.0116 (0.0134)	−9.126 (6.923)
7 weeks since COVID-19 outbreak	0.0419 (0.0360)	0.0122 (0.0135)	−0.847 (7.067)
8 weeks since COVID-19 outbreak	0.0820** (0.0357)	0.0309** (0.0134)	−7.239 (6.790)
9 weeks since COVID-19 outbreak	0.173*** (0.0349)	0.0883*** (0.0137)	−3.630 (6.236)
10 weeks since COVID-19 outbreak	0.169*** (0.0348)	0.0729*** (0.0133)	−13.69** (6.407)
11 weeks since COVID-19 outbreak	0.126*** (0.0358)	0.0627*** (0.0137)	−8.390 (6.604)
12 weeks since COVID-19 outbreak	0.183*** (0.0350)	0.0899*** (0.0136)	4.469 (6.549)
13 weeks since COVID-19 outbreak	0.117*** (0.0357)	0.0617*** (0.0137)	−3.955 (6.112)
Region controls: weather, cram school closure	Yes	Yes	Yes
Observations	615,249	615,249	225,369

Notes. Robust standard errors are in parentheses. The dependent variables are (a) the log number of searches (i.e., quantity), (b) the diversity index of weekly searches (i.e., pattern), and (c) the curriculum index (i.e., pace). The estimator is OLS with individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table E.3. Results with Additional Dynamic Region-Specific Controls: Employment and Retail Sales

Dependent variable	Quantity	Pattern	Pace
9 weeks until COVID-19 outbreak	−0.0355 (0.0271)	−0.00144 (0.00995)	−7.455 (7.261)
8 weeks until COVID-19 outbreak	0.0193 (0.0261)	0.0152 (0.00960)	−9.474 (7.261)
7 weeks until COVID-19 outbreak	0.0192 (0.0268)	0.00748 (0.00998)	5.633 (6.955)
6 weeks until COVID-19 outbreak	0.0523* (0.0268)	0.0209** (0.0104)	−0.590 (6.049)
5 weeks until COVID-19 outbreak	0.0186 (0.0259)	0.0231** (0.0102)	−10.70* (5.582)
4 weeks until COVID-19 outbreak	0.0197 (0.0244)	0.0110 (0.00948)	−3.217 (5.637)
3 weeks until COVID-19 outbreak	0.0355 (0.0240)	0.0158* (0.00934)	−5.296 (5.769)
2 weeks until COVID-19 outbreak	−0.0250 (0.0204)	−0.00770 (0.00849)	−1.117 (5.213)
1 week until COVID-19 outbreak		Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.0973*** (0.0220)	−0.0478*** (0.00913)	−3.914 (5.586)
1 week since COVID-19 outbreak	−0.0739*** (0.0273)	0.0108 (0.0105)	−20.97*** (6.621)
2 weeks since COVID-19 outbreak	−0.0358 (0.0287)	0.0139 (0.0109)	−14.29** (6.399)
3 weeks since COVID-19 outbreak	−0.0618** (0.0304)	−0.00832 (0.0113)	−11.00 (6.780)
4 weeks since COVID-19 outbreak	−0.0652** (0.0316)	−0.00534 (0.0118)	−3.517 (6.940)
5 weeks since COVID-19 outbreak	0.00953 (0.0319)	0.0165 (0.0119)	−5.288 (6.318)
6 weeks since COVID-19 outbreak	0.0165 (0.0326)	0.0271** (0.0121)	−9.186 (6.098)
7 weeks since COVID-19 outbreak	0.0631** (0.0321)	0.0292** (0.0120)	−2.005 (6.204)
8 weeks since COVID-19 outbreak	0.0909*** (0.0325)	(a)0.0429*** (0.0122)	−8.386 (6.152)
9 weeks since COVID-19 outbreak	0.169*** (0.0332)	0.0852*** (0.0131)	−7.768 (5.836)
10 weeks since COVID-19 outbreak	0.162*** (0.0331)	0.0737*** (0.0127)	−15.53** (6.061)
11 weeks since COVID-19 outbreak	0.112*** (0.0329)	0.0649*** (0.0127)	−11.98** (6.110)
12 weeks since COVID-19 outbreak	0.150*** (0.0334)	0.0774*** (0.0130)	−0.856 (6.196)
13 weeks since COVID-19 outbreak	0.0836** (0.0332)	0.0522*** (0.0128)	−8.678 (5.716)
Region controls: weather, employment, and retail sales	Yes	Yes	Yes
Observations	828,000	828,000	322,943

Notes. Robust standard errors are in parentheses. The dependent variables are (a) the log number of searches (i.e., quantity), (b) the diversity index of weekly searches (i.e., pattern), and (c) the curriculum index (i.e., pace). The estimator is OLS with individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table E.4. Results with Additional Dynamic Region-Specific Controls: Cram School Closure Rates, Employment, and Retail Sales

Dependent variable	Quantity	Pattern	Pace
9 weeks until COVID-19 outbreak	−0.0274 (0.0274)	0.000648 (0.0100)	−4.064 (7.280)
8 weeks until COVID-19 outbreak	0.0300 (0.0266)	0.0172* (0.00977)	−5.001 (7.319)
7 weeks until COVID-19 outbreak	0.0293 (0.0274)	0.00913 (0.0102)	9.605 (7.015)
6 weeks until COVID-19 outbreak	0.0593** (0.0275)	0.0219** (0.0106)	2.565 (6.129)
5 weeks until COVID-19 outbreak	0.0230 (0.0268)	0.0242** (0.0105)	−7.102 (5.691)
4 weeks until COVID-19 outbreak	0.0230 (0.0249)	0.0110 (0.00966)	−1.138 (5.704)
3 weeks until COVID-19 outbreak	0.0282 (0.0245)	0.0116 (0.00952)	−4.735 (5.866)
2 weeks until COVID-19 outbreak	−0.0243 (0.0205)	−0.00773 (0.00852)	0.144 (5.216)
1 week until COVID-19 outbreak		Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.119*** (0.0222)	−0.0546*** (0.00923)	−7.199 (5.642)
1 week since COVID-19 outbreak	−0.0867*** (0.0304)	−0.00193 (0.0116)	−21.08*** (7.127)
2 weeks since COVID-19 outbreak	−0.0531* (0.0319)	0.00128 (0.0120)	−15.98** (6.972)
3 weeks since COVID-19 outbreak	−0.0817** (0.0330)	−0.0211* (0.0122)	−14.78** (7.267)
4 weeks since COVID-19 outbreak	−0.104*** (0.0338)	−0.0193 (0.0126)	−7.777 (7.405)
5 weeks since COVID-19 outbreak	−0.0452 (0.0366)	−0.0110 (0.0136)	−9.257 (7.240)
6 weeks since COVID-19 outbreak	−0.0398 (0.0365)	0.00191 (0.0135)	−11.72* (6.962)
7 weeks since COVID-19 outbreak	0.0132 (0.0361)	0.00259 (0.0135)	−3.331 (7.057)
8 weeks since COVID-19 outbreak	0.0621* (0.0358)	0.0238* (0.0134)	−8.884 (6.787)
9 weeks since COVID-19 outbreak	0.156*** (0.0349)	0.0824*** (0.0137)	−5.134 (6.229)
10 weeks since COVID-19 outbreak	0.155*** (0.0348)	0.0678*** (0.0133)	−14.94** (6.403)
11 weeks since COVID-19 outbreak	0.119*** (0.0358)	0.0594*** (0.0137)	−8.954 (6.620)
12 weeks since COVID-19 outbreak	0.175*** (0.0351)	0.0861*** (0.0136)	3.505 (6.569)
13 weeks since COVID-19 outbreak	0.116*** (0.0359)	0.0603*** (0.0138)	−4.076 (6.146)
Region controls: weather, cram school closure, economic factors	Yes	Yes	Yes
Observations	615,249	615,249	225,369

Notes. Robust standard errors are in parentheses. The dependent variables are (a) the log number of searches (i.e., quantity), (b) the diversity index of weekly searches (i.e., pattern), and (c) the curriculum index (i.e., pace). The estimator is OLS with individual and time fixed effects.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

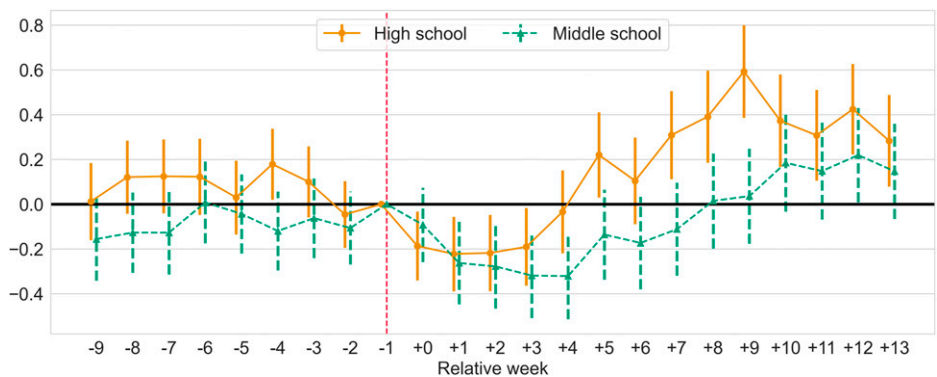
Table E.5. Subsample Analysis with Only Students Who Have Used Other QANDA Learning Features Along with the Main Search Service

Dependent variable	Quantity	Pattern	Pace
9 weeks until COVID-19 outbreak	−0.0219 (0.0437)	0.00192 (0.0163)	−3.815 (8.448)
8 weeks until COVID-19 outbreak	0.0622 (0.0416)	0.0342** (0.0159)	0.424 (8.088)
7 weeks until COVID-19 outbreak	0.0221 (0.0424)	−0.00116 (0.0162)	9.457 (7.898)
6 weeks until COVID-19 outbreak	0.0565 (0.0418)	0.0280* (0.0167)	1.326 (6.927)
5 weeks until COVID-19 outbreak	0.0379 (0.0408)	0.0233 (0.0163)	−9.460 (6.034)
4 weeks until COVID-19 outbreak	0.0267 (0.0382)	0.0132 (0.0155)	−5.305 (6.311)
3 weeks until COVID-19 outbreak	0.0456 (0.0381)	0.00876 (0.0154)	−2.114 (6.219)
2 weeks until COVID-19 outbreak	−0.0698** (0.0321)	−0.0342** (0.0141)	−2.762 (5.620)
1 week until COVID-19 outbreak		Omitted (baseline)	
0 weeks since COVID-19 outbreak	−0.151*** (0.0353)	−0.0806*** (0.0153)	−3.438 (6.223)
1 week since COVID-19 outbreak	−0.150*** (0.0439)	0.000591 (0.0173)	−25.75*** (7.293)
2 weeks since COVID-19 outbreak	−0.110** (0.0452)	0.000294 (0.0177)	−20.54*** (7.112)
3 weeks since COVID-19 outbreak	−0.172*** (0.0467)	−0.0337* (0.0180)	−15.80** (7.716)
4 weeks since COVID-19 outbreak	−0.189*** (0.0470)	−0.0403** (0.0181)	−17.85** (7.731)
5 weeks since COVID-19 outbreak	−0.114** (0.0489)	−0.0183 (0.0185)	−23.20*** (6.963)
6 weeks since COVID-19 outbreak	−0.0628 (0.0506)	0.00761 (0.0190)	−25.37*** (6.924)
7 weeks since COVID-19 outbreak	−0.000944 (0.0496)	0.0112 (0.0189)	−12.64* (7.157)
8 weeks since COVID-19 outbreak	0.0212 (0.0505)	0.0223 (0.0191)	−15.88** (7.163)
9 weeks since COVID-19 outbreak	0.0671 (0.0508)	0.0569*** (0.0202)	−17.66** (6.930)
10 weeks since COVID-19 outbreak	0.113** (0.0510)	0.0631*** (0.0199)	−18.89** (7.445)
11 weeks since COVID-19 outbreak	0.111** (0.0514)	0.0678*** (0.0200)	−19.61*** (7.351)
12 weeks since COVID-19 outbreak	0.159*** (0.0514)	0.0882*** (0.0201)	−5.344 (7.516)
13 weeks since COVID-19 outbreak	0.105** (0.0520)	0.0615*** (0.0200)	−16.89** (6.780)
Region controls: weather	Yes	Yes	Yes
Observations	404,064	404,064	201,918

Notes. Robust standard errors are in parentheses. The dependent variables are (a) the log number of searches (i.e., quantity), (b) the diversity index of weekly searches (i.e., pattern), and (c) the curriculum index (i.e., pace). The estimator is OLS with individual and time fixed effects.

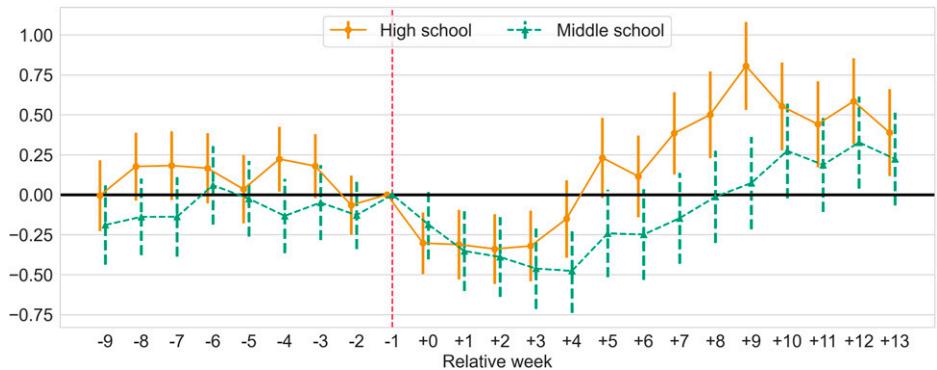
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure E.1. (Color online) Results with Average Time Consumption (per Problem) as the Dependent Variable



Notes. Vertical bars denote 95% confidence intervals. The dependent variable is the average time per problem (in seconds) of reviewing problem solutions. The estimator is OLS with individual and time fixed effects. The dummy variable indicating a week before treatment in DG is omitted as the base group and denoted as a dashed vertical line.

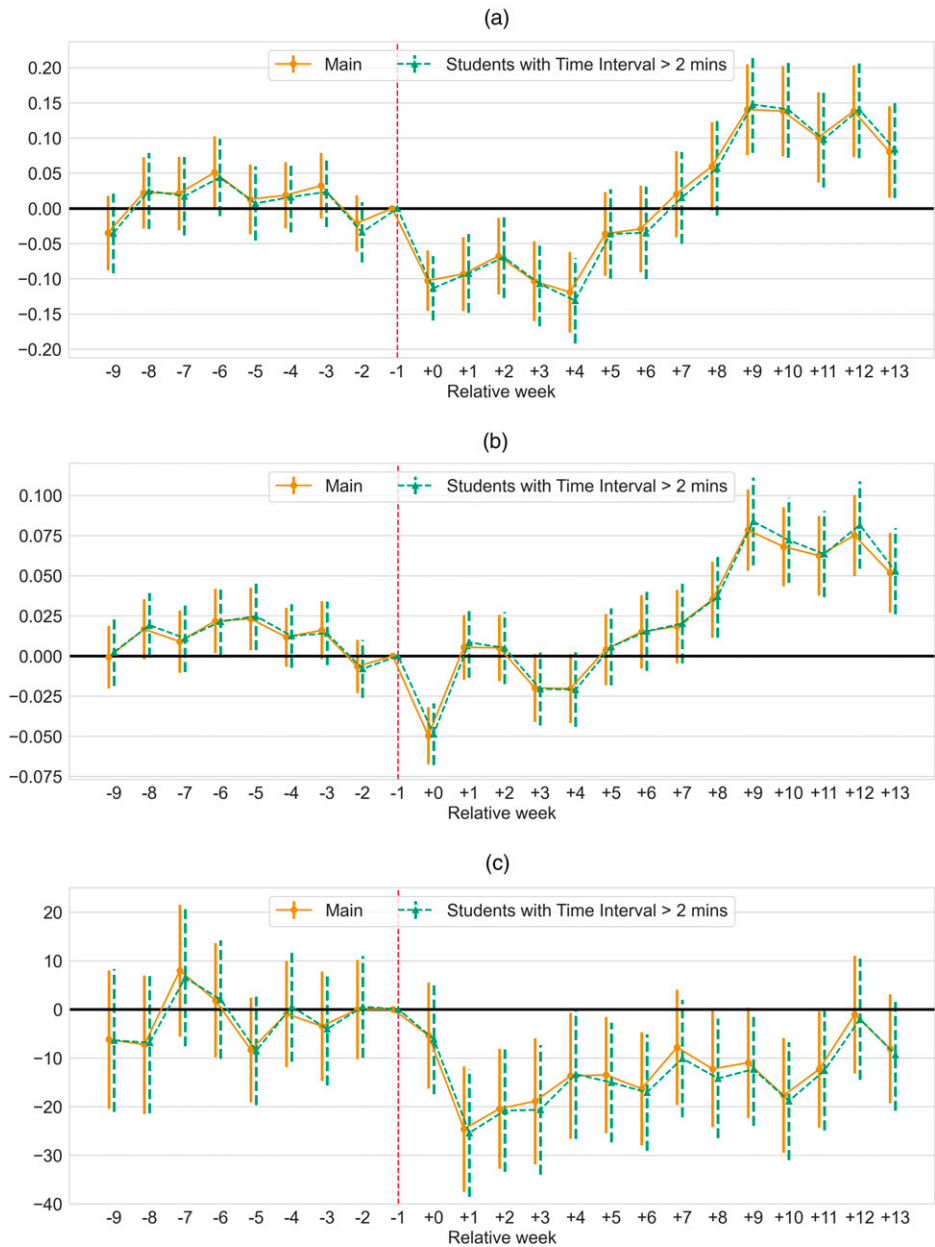
Figure E.2. (Color online) Results with Total Time Consumption as the Dependent Variable



Notes. Vertical bars denote 95% confidence intervals. The dependent variable is the total time (in seconds) of reviewing problem solutions. The estimator is OLS with individual and time fixed effects. The dummy variable indicating a week before treatment in DG is omitted as the base group and denoted as a dashed vertical line.

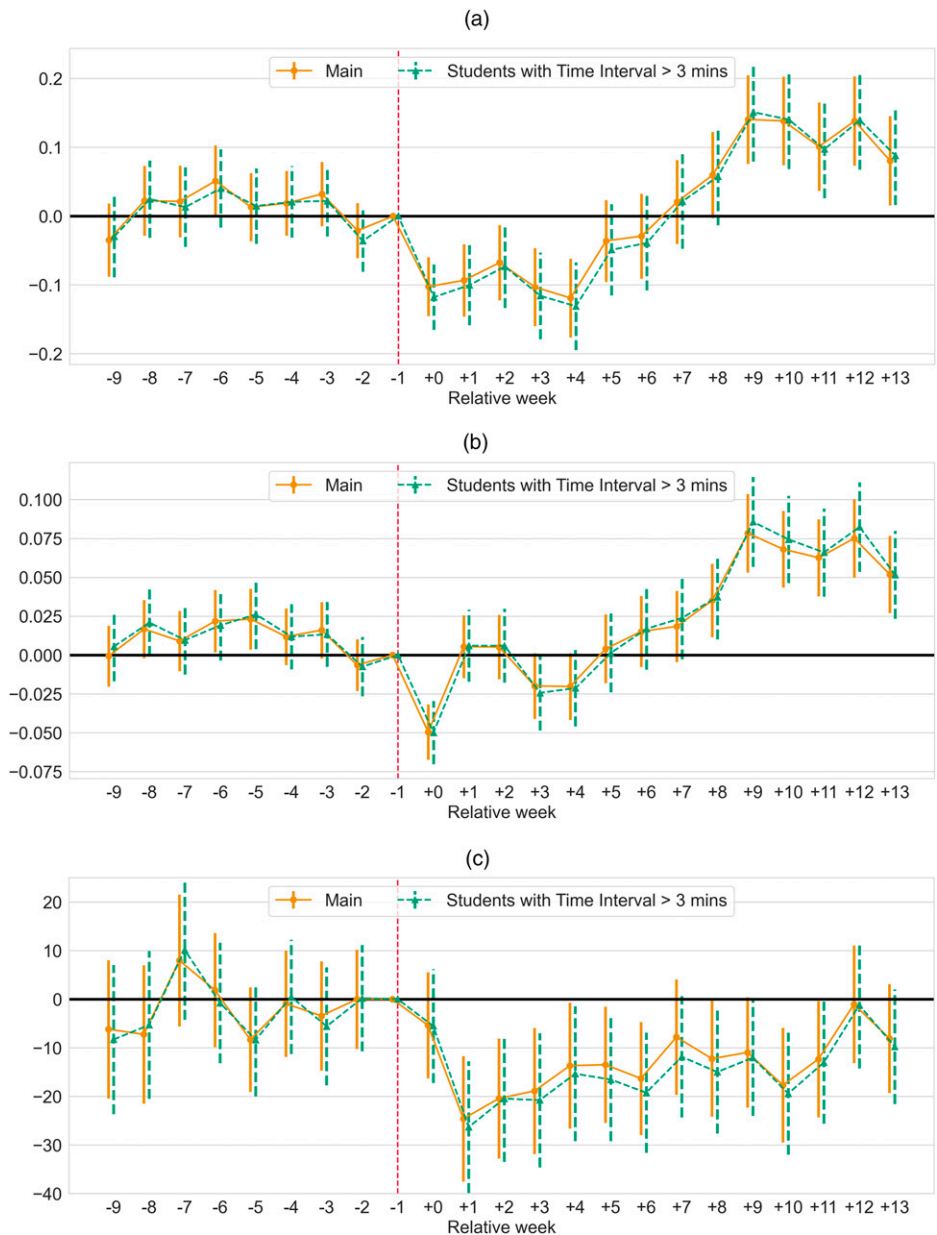
Appendix F. Alternative Purposes of Using the App

Figure F.1. (Color online) All (36,000) vs. Subsample of Students with a Time Interval of 2+ Minutes (32,228)



Notes. (a) Quantity. (b) Pattern. (c) Pace. Vertical bars denote 95% confidence intervals. The dependent variables are (a) the log number of searches, (b) the diversity index of weekly searches, and (c) the curriculum index. The estimator is OLS with individual and time fixed effects. The dummy variable indicating a week before treatment in DG is omitted as the base group and denoted as a dashed vertical line.

Figure F.2. (Color online) All (36,000) vs. Subsample of Students with a Time Interval of 3+ Minutes (29,155)



Notes. (a) Quantity. (b) Pattern. (c) Pace. Vertical bars denote 95% confidence intervals. The dependent variables are (a) the log number of searches, (b) the diversity index of weekly searches, and (c) the curriculum index. The estimator is OLS with individual and time fixed effects. The dummy variable indicating a week before treatment in DG is omitted as the base group and denoted as a dashed vertical line.

Endnotes

¹ When referring to ICT in this paper, ICT denotes QANDA, the AI-driven math education platform. We reserve more detailed information on QANDA subsequently in Section 3.1.
² <https://classroom.google.com>.
³ <https://www.freckle.com/math>.
⁴ <https://zoom.us>.
⁵ <https://photomath.app>.
⁶ <https://math.microsoft.com>.
⁷ Korea Ministry of Education Bulletin No. 2015-74Ho (Supplement 8), <https://www.moe.go.kr/boardCnts/view.do?boardID=141&boardSeq=60747&lev=0> (accessed June 28, 2022).

⁸ The three services are ipinfo.io, ipdata.co, and getgeoapi.com.
⁹ See <https://edition.cnn.com/2020/02/26/asia/shincheonji-south-korea-hnk-intl/> (accessed June 28, 2022).
¹⁰ The estimation result tables are provided in Appendix D. We confirm that the parallel-trend assumption holds, as only a few pre-treatment time dummies in the model are significant.
¹¹ Our baseline period is one week before (–1) the outbreak. As such, the effect of pandemic threat on search volume in the ninth week after (+9) the outbreak is calculated by comparing the changes in search volume from one week before (–1) to the ninth week following (+9) the outbreak between the DG and non-DG regions. Although cram-school closure rates in both regions were at similar levels before the outbreak, it remained relatively higher at 37% in

DG compared with non-DG (e.g., 14% in Seoul) at the ninth week after the outbreak. If QANDA is a complement for cram schools and it is the sole factor explaining QANDA use, the effect of pandemic threat on search volume should be negative at the ninth week after the pandemic. However, this is not the case, as shown by the data and empirical results.

¹² We highlight that cram-school closure rate is quite sparse and incomplete because it is not one of the key statistics collected or monitored on a regular basis by a centralized institution and has been reported largely ad hoc. For example, local government agencies have tracked cram-school closures over certain time periods, owing mostly to the pandemic. We collected as much information about cram-school closure rates as possible by directly contacting the Office of Education in each of the 17 regions in South Korea (seven metropolitan cities, one special self-governing city, and nine provinces). In the end, we were able to obtain cram-school closure rates for 25.1% of all region-week combinations. After controlling for cram-school closure rates, the pace (i.e., curriculum index) recovers to the prepandemic level starting in the third week (+3) after the outbreak, whereas the main analysis shows that it starts to recover in the seventh week (+7). As we include an additional control, a few marginally significant estimates become insignificant. Also, the sample size is different due to the limitation of cram-school data. However, the overall trend of recovery remains consistent with the main results.

¹³ We thank an anonymous reviewer for suggesting this robustness check.

¹⁴ The explanatory variables in the first stage include (a) the number of nonsearch activities performed using QANDA (e.g., watching lecture videos, solving problem sets, and reading concept books), (b) the lag of the number of searches, (c) individual characteristics (e.g., grade and region), and (d) weather information (e.g., weekly average temperature, the number of days with rain per week, and the number of days with snow per week). We assert that nonsearch activities and the number of searches in the previous week affect the decision to search this week (i.e., complementary usage or inertia), but not the content to study. So, we satisfy the exclusion restriction by including variables (a) and (b) in the first stage (i.e., the selection equation), but not in the second stage (i.e., the outcome equation).

¹⁵ <https://kostat.go.kr>; South Korea's counterpart to the U.S. Census Bureau. We note that the finest time granularity provided by Statistics Korea is at the monthly level. Results are robust to different interpolation and imputation approaches for converting the monthly frequency of these statistics to weekly.

¹⁶ We note that our results are largely consistent when including all region-specific, time-varying controls discussed in the paper (i.e., weather-related, cram-school closure, and economic variables). See Table E.4 in Appendix E. There is a three- or four-period difference in the starting point of the recovery in terms of pace between the main analysis and the robustness analysis with more region-specific, time-varying controls. This is because a few marginally significant estimates become insignificant as more control variables are included. However, the overall trend of recovery remains largely consistent with the main results.

¹⁷ We thank an anonymous reviewer for suggesting this section.

¹⁸ <https://www.whitehouse.gov/build-back-better/>.

¹⁹ <https://trends.google.com>. We note that Google Trends provides only relative search volumes, not their absolute raw search intensity. Given that both population and Google's search market share in South Korea have largely held steady during our data period (according to Internet Trend; <http://www.internettrend.co.kr>), we used the population of regions to make the relative search volumes comparable across regions.

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